Python & Data Science Notes

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1. Environment Setup

It is always a good practice to start with creating a separate environment.

Three Ways to Create Virtual Environments

1. Using `venv` Module

```
python -m venv venv_name
venv_name\Scripts\activate      # Activate
deactivate                       # Deactivate
```

2. Using `virtualenv` (Linux / PowerShell / Git Bash)

```
pip install virtualenv
virtualenv -p python3 venv_name
venv_name\Scripts\activate      # Activate
deactivate                       # Deactivate
```

3. Using Anaconda *(Recommended)*

```
conda create -p venv_name python==3.14 -y    # Can use other versions
conda activate venv/                          # Activate
conda deactivate                              # Deactivate
```

Setting Up with Anaconda (Step-by-Step)

- Open command prompt / terminal in VS Code
- Run: `conda create -p venv python==3.14`
- Type `y` to confirm → environment created
- Add a `requirements.txt` file
- Add package names inside `requirements.txt`
- Activate: `conda activate venv/`
- Install packages from file: `pip install -r requirements.txt`
- **Jupyter Notebook** (`.ipynb`) requires two packages: `ipykernel` + Jupyter extension
- Click on "detecting kernels" in `.ipynb` file and select the created environment

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2. Python Modules and Packages

Helps organize and reuse code.

Importing Modules

```
# Way 1 — import the whole module
import math
math.sqrt(16)

# Way 2 — import specific functions
from math import sqrt, pi
print(sqrt(16))
print(sqrt(25))
print(pi)

# Importing everything
from math import *
print(sqrt(16))
print(pi)

# Using an alias
import numpy as np     # np is the alias
np.array([1, 2, 3])    # access numpy with np.
```

Creating Your Own Custom Module / Package

- Make a folder (e.g. `package/`)
- Inside it, create a special file: **`__init__.py`** (makes Python treat the folder as a package)
- Create your module file inside: `package/maths.py`
- Define your functions inside that file

```
# Calling your custom module
from package.maths import addition
addition(2, 3)

# OR
from package import maths
maths.addition(2, 3)
```

Syntax:

```
from folder_name.package_name import function_name
```

> **Note:** For subpackages (nested folders), place an `__init__.py` in each subfolder too.

```
from folder_name.subpackageFolder_name.packagename import function_name
function_name()
```

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3. Python Standard Library Overview

In Python, the standard library is a big collection of built-in modules that come with Python itself — no extra installs needed. Think of it as Python's **"batteries included"** toolbox.

`array` — Basic Arrays

```
import array
arr = array.array('i', [1, 2, 3, 4])
print(arr)
```

`math` — Mathematical Functions

```
import math
math.sqrt(16)
```

`random` — Random Values

```
import random
print(random.randint(1, 10))
print(random.choice(['apple', 'banana', 'cherry']))
```

`os` — Operating System / IO Operations

```
import os
print(os.getcwd())
os.mkdir('test_dir')
```

`shutil` — High-Level File Operations

```
import shutil
shutil.copyfile('source.txt', 'destination.txt')
```

`json` — Data Serialization (used in APIs)

```
import json

data = {'name': 'Sher', 'age': 25}
json_str = json.dumps(data)        # Dict → JSON string
```

```
print(json_str)
print(type(json_str))

parsed_data = json.loads(json_str) # JSON string → Dict
print(parsed_data)
print(type(parsed_data))
```

Result:

```
{"name": "Sher", "age": 25}
<class 'str'>
{'name': 'Sher', 'age': 25}
<class 'dict'>
```

`csv` — Working with CSV Files

```
import csv

with open('example.csv', mode='w', newline='') as file:
    writer = csv.writer(file)
    writer.writerow(['name', 'age'])
    writer.writerow(['Sher', '20'])

with open('example.csv', mode='r') as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)
```

`datetime` — Date and Time

```
from datetime import datetime, timedelta    # timedelta is used for time changes

now = datetime.now()
print(now)

yesterday = now - timedelta(days=1)
print(yesterday)
```

`time` — Time / Sleep

```
import time
print(time.time())
```

```
time.sleep(2)          # Program sleeps for 2 seconds
print(time.time())
```

`re` — Regular Expressions

`re` lets you search, match, split, and replace text using patterns (regex).

```
import re

pattern = r'\d+'
text = 'There are 212 apples'
match = re.search(pattern, text)
print(match.group())
```

Result:

```
212
```

F-Strings

```
x = 10
print(f"my score is {x}")
```

---

4. Working with File Paths

Listing Directory Contents

```
import os
items = os.listdir('.')
print(items)
```

Joining Paths

```
import os

dir_name = "folder"
file_name = "file.txt"
full_path = os.path.join(dir_name, file_name)
full_final_path = os.path.join(os.getcwd(), dir_name, file_name)
print(full_path)
print(full_final_path)
```

Checking if a File Exists

```
import os

path = 'example1.txt'
if os.path.exists(path):
    print(f"The path '{path}' exists")
else:
    print(f"The path '{path}' does not exist")
```

Checking if Path is a File or Directory

```
import os

path = 'Notes.txt'
if os.path.isfile(path):
    print(f"'{path}' is a file")
elif os.path.isdir(path):
    print(f"'{path}' is a directory")
else:
    print(f"'{path}' is neither a directory nor a file")
```

Getting the Absolute Path

```
relative_path = 'Notes.txt'
absolute_path = os.path.abspath(relative_path)
print(absolute_path)
```

> **Note:** This is better than manually joining with `os.path.join(os.getcwd(), ...)`.

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5. Exception Handling

Basic `try-except` Block

```
try:
    a = b
except:
    print("Variable not assigned")
```

Displaying the Error Type

```
try:
    a = b
except NameError as ex:
    print(ex)
```

Syntax:

```
try:
    # code
except ErrorType as alias:
    print(alias)
```

Catching Specific Exceptions

```
# ZeroDivisionError
try:
    result = 1 / 0
except ZeroDivisionError as err:
    print("Enter denom greater than 0")
```

Multiple `except` Blocks

> **Note:** All error types derive from the base `Exception` class — always put `Exception` **last**.

```
try:
    numb = int(input("Enter a value: "))
    res = 10 / numb
except ValueError:
    print("Enter valid value")
except ZeroDivisionError:
    print("Enter greater than 0")
```

```
except Exception as ex:
    print(ex)
```

`try`, `except`, and `else`

`else` executes **only if** `try` runs successfully without triggering any `except` block.

```
try:
    numb = int(input("Enter a value: "))
    res = 10 / numb
except ValueError:
    print("Enter valid value")
except ZeroDivisionError:
    print("Enter greater than 0")
except Exception as ex:
    print(ex)
else:
    print(f"The value is {res}")
```

`try`, `except`, `else`, and `finally`

`finally` executes **no matter what** — whether the `try` succeeded or an exception was caught.

```
try:
    numb = int(input("Enter a value: "))
    res = 10 / numb
except ValueError:
    print("Enter valid value")
except ZeroDivisionError:
    print("Enter greater than 0")
except Exception as ex:
    print(ex)
else:
    print(f"The value is {res}")
finally:
    print("Execution completed")
```

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6. NumPy

NumPy is a fundamental library for scientific computing in Python. It provides support for arrays and matrices, along with a collection of mathematical functions to operate on these data structures.

```
import numpy as np
```

Creating Arrays

1D Array

```
arr1 = np.array([1, 2, 3, 4, 5])
print(arr1)
print(type(arr1))       # Not a list!
print(arr1.shape)       # (5,) — no rows/columns in 1D
```

Reshaping a 1D Array to 2D

```
arr2 = np.array([1, 2, 3, 4, 5])
arr2.reshape(1, 5)      # 1 row, 5 columns → array([[1, 2, 3, 4, 5]])
```

> Two `[[]]` brackets = 2D array, three `[[[]]]` = 3D array.

2D and 3D Arrays

```
arr3 = np.array([[1, 2, 3, 4, 5]])           # 2D — one row
arr4 = np.array([[1, 2, 3, 4], [5, 6, 7, 8]])  # 2D — two rows
print(arr4.shape)    # (2, 4) → 2 rows, 4 columns
```

Inbuilt Array Creation Functions

```
np.arange(0, 10, 2)              # [0, 2, 4, 6, 8]
np.arange(0, 10, 2).reshape(5, 1)  # Reshape to 5 rows, 1 column

np.ones((3, 4))    # 3x4 array of 1s
np.eye(3)          # 3x3 identity matrix
```

Array Attributes

```
arr = np.array([[1, 2, 3], [4, 5, 6]])

print("Array:\n", arr)
print("Shape:", arr.shape)                    # (2, 3)
print("Number of dimensions:", arr.ndim)      # 2
print("Size (number of elements):", arr.size) # 6
print("Data type:", arr.dtype)                # int64
print("Item size (in bytes):", arr.itemsize)  # 8
```

Vectorized Operations

```
arr1 = np.array([1, 2, 3, 4, 5])
arr2 = np.array([10, 20, 30, 40, 50])

print("Addition:", arr1 + arr2)
print("Subtraction:", arr2 - arr1)
# Similarly: multiplication, division, etc.
```

Universal Functions

Functions that apply to the **entire array** at once.

```
arr = np.array([2, 3, 4, 5, 6])

print(np.sqrt(arr))   # Square root
print(np.exp(arr))    # Exponential
print(np.sin(arr))    # Sine
print(np.log(arr))    # Natural log
```

Array Slicing and Indexing

```
arr = np.array([[1, 2, 3, 4],
                [5, 6, 7, 8],
                [9, 10, 11, 12]])

arr[0][0]          # Single element → 1
arr[1:, 2:]        # Pick 7, 8, 11, 12
arr[0:2, 2:4]      # Pick 3, 4, 7, 8
```

Modifying Array Elements

```
arr[0, 0] = 100        # Modify single element
arr[1:] = 99           # Set all rows from index 1 to 99
```

```
arr[1:3, 2:4] = 34     # Modify a slice
```

Statistical Concepts — Normalization

```
data = np.array([1, 2, 3, 4, 5])

mean = np.mean(data)
std_dev = np.std(data)
normalized_data = (data - mean) / std_dev

print(normalized_data)
```

Statistical Operations

```
data = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])

print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Standard Deviation:", np.std(data))
print("Variance:", np.var(data))
```

Logical Operations

```
data = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])

data[(data >= 5) & (data <= 8)]    # Elements between 5 and 8
# Any logical operation can be used: &, |, ~
```

---

7. Pandas — Data Manipulation Library

Pandas is a powerful data manipulation library in Python, widely used for data analysis and cleaning. It provides two primary data structures:

- **Series** — One-dimensional array-like object
- **DataFrame** — Two-dimensional, size-mutable, tabular data structure with labeled axes (rows and columns)

```
import pandas as pd
```

7.1 Pandas Series

A Pandas Series is a one-dimensional array-like object that can hold any data type. Similar to a column in a table.

From a List

```
data = [1, 2, 3, 4, 5]
series = pd.Series(data)
print("Series \n", series)
print(type(series))
```

Result:

```
0    1
1    2
2    3
3    4
4    5
dtype: int64
<class 'pandas.core.series.Series'>
```

From a Dictionary

```
data = {'a': 1, 'b': 2, 'c': 3}
series_dict = pd.Series(data)
print(series_dict)
```

Result:

```
a    1
b    2
c    3
dtype: int64
```

> Keys become the index, values are the series values.

With a Custom Index

```
data = [10, 20, 30]
index = ['a', 'b', 'c']
pd.Series(data, index=index)
```

> **Result:**
>
> ```
> a    10
> b    20
> c    30
> dtype: int64
> ```

7.2 Pandas DataFrame

A Pandas DataFrame is a two-dimensional, size-mutable, tabular data structure with labeled axes.

Reading Data

Method	Use
`pd.read_csv()`	CSV or delimited text files
`pd.read_excel()`	`.xlsx` / `.xls` files (supports multiple sheets)
`pd.read_json()`	Web-based JSON data
`pd.read_sql()`	SQL databases (PostgreSQL, MySQL, SQLite)

Creating a DataFrame from a Dictionary of Lists

```
data = {
    'Name': ['Sher', 'Hassan', 'Khan'],
    'Age':  [18, 19, 20],
    'City': ['Quetta', 'Islamabad', 'Zhob']
}
```

```
df = pd.DataFrame(data)
print(df)
```

Result:

```
      Name  Age       City
0     Sher   18     Quetta
1   Hassan   19  Islamabad
2     Khan   20       Zhob
```

Creating a DataFrame from a List of Dictionaries

```
data1 = [
    {'Name': 'Sher',   'Age': 20, 'City': 'Quetta'},
    {'Name': 'Hassan', 'Age': 20, 'City': 'Islamabad'},
    {'Name': 'Khan',   'Age': 20, 'City': 'Zhob'},
    {'Name': 'Ali',    'Age': 20, 'City': 'Karachi'}
]
df1 = pd.DataFrame(data1)
print(df1)
```

Result:

```
      Name  Age       City
0     Sher   20     Quetta
1   Hassan   20  Islamabad
2     Khan   20       Zhob
3      Ali   20    Karachi
```

Reading a CSV File

```
df = pd.read_csv('sales_data.csv')
print(df.head(5))    # First 5 rows
print(df.tail(5))    # Last 5 rows
```

7.3 Accessing Data from a DataFrame

```
df['Name']              # Access a column (returns Series)
df.loc[0]               # Access row by label/index
df.iloc[0]              # Access row by position
df.loc[0, 'Name']       # Specific element by label
```

```
df.iloc[0, 2]           # Specific element by position
df.at[1, 'Age']         # Element with .at (label-based)
df.iat[1, 2]            # Element with .iat (position-based)
```

7.4 Data Manipulation with DataFrame

Adding a New Column

```
df['Salary'] = [50000, 60000, 70000, 80000]
```

Removing a Column

```
df.drop('Salary', axis=1)             # axis=1 for columns
df.drop(columns=['Salary'])           # Alternative
df = df.drop(columns=['Salary'])      # Save changes
df.drop('Salary', axis=1, inplace=True)  # In-place
```

> **Note:** Default `axis=0` removes rows, not columns!

Modifying Column Values

```
df['Age'] = df['Age'] + 1
```

Removing a Row

```
df.drop(3, inplace=True)    # Remove row at index 3
```

Statistical Summary

```
df.describe()
```

> **Result:**
>
> ```
>  Age    Salary
> count    3.0     3.0
> mean    21.0  60000.0
> std      1.0  10000.0
> ```

```
min     20.0  50000.0
25%     20.5  55000.0
50%     21.0  60000.0
75%     21.5  65000.0
max     22.0  70000.0
```

7.5 Data Manipulation and Analysis

Checking Data Types

```
df.dtypes
```

Handling Missing Values

```
df.isnull()               # True where entries are missing
df.isnull().any(axis=1)   # Which rows have missing values
df.isnull().sum()         # Count of missing values per column
```

Result:

```
Unit Price    4
...all others: 0
```

Filling Missing Values

```
df.fillna(0)    # Fill all NaN with 0

# Fill with the column mean
df['Unit_Price_fillNA'] = df['Unit Price'].fillna(df['Unit Price'].mean())
```

Renaming Columns — `.rename()`

```
# Syntax: df.rename(columns={'OldName': 'NewName'})
df = df.rename(columns={'Country': 'Nationality'})
```

Changing Data Types — `.astype()`

```
# Convert Units Sold from int to float
df['New Units Sold'] = df['Units Sold'].astype(float)
```

Applying Functions — `.apply()`

```
df['New Value'] = df['New Units Sold'].apply(lambda x: x * 2)
```

7.6 Data Aggregation and Grouping

Group by One Column

```
grouped_mean = df.groupby('Region')['Unit Cost'].mean()
print(grouped_mean)
```

Result:

```
Region
Asia                                239.587273
Australia and Oceania               154.744545
Central America and the Caribbean   157.817143
Europe                              223.166364
Middle East and North Africa        152.450000
North America                       205.293333
Sub-Saharan Africa                  183.677500
Name: Unit Cost, dtype: float64
```

Group by Two Columns

```
grouped_sum = df.groupby(['Region', 'Item Type'])['Unit Cost'].sum()
print(grouped_sum)
```

Aggregate Multiple Functions

```
grouped_agg = df.groupby('Region')['Unit Cost'].agg(['mean', 'sum', 'count'])
print(grouped_agg)
```

Result:

```
                                        mean       sum  count
Region
Asia                                239.587273   2635.46     11
Australia and Oceania               154.744545   1702.19     11
Central America and the Caribbean   157.817143   1104.72      7
Europe                              223.166364   4909.66     22
Middle East and North Africa        152.450000   1524.50     10
North America                       205.293333    615.88      3
Sub-Saharan Africa                  183.677500   6612.39     36
```

7.7 Merging and Joining DataFrames

```
df1 = pd.DataFrame({'Key': ['A', 'B', 'C'], 'Value1': [1, 2, 3]})
df2 = pd.DataFrame({'Key': ['A', 'B', 'D'], 'Value1': [4, 5, 6]})
```

Inner Join — only matching keys

```
pd.merge(df1, df2, on="Key", how="inner")
```

Result:

```
  Key  Value1_x  Value1_y
0   A         1         4
1   B         2         5
```

Outer Join — all keys, NaN where no match

```
pd.merge(df1, df2, on="Key", how="outer")
```

Result:

```
  Key  Value1_x  Value1_y
0   A       1.0       4.0
1   B       2.0       5.0
2   C       3.0       NaN
3   D       NaN       6.0
```

Left Join — all keys from left DataFrame

```
pd.merge(df1, df2, on="Key", how="left")
```

Result:

```
  Key  Value1_x  Value1_y
0   A         1       4.0
1   B         2       5.0
2   C         3       NaN
```

Right Join — all keys from right DataFrame

```
pd.merge(df1, df2, on="Key", how="right")
```

Result:

```
  Key  Value1_x  Value1_y
0   A       1.0         4
1   B       2.0         5
2   D       NaN         6
```

---

8. Reading Data From Different Sources

8.1 Reading Data from JSON — `.read_json()` and `.StringIO()`

The flow is: **Read Data From JSON → Convert to String → Store**

```
import pandas as pd
from io import StringIO

Data = '{"employee_name": "James", "email": "james@gmail.com", "job_profile":
[{"title1":"Team Lead"}, {"title2": "Sr. Developer"}]}'
df = pd.read_json(StringIO(Data))
df
```

Result:

```
  employee_name            email              job_profile
0         James  james@gmail.com   {'title1': 'Team Lead'}
1         James  james@gmail.com  {'title2': 'Sr. Developer'}
```

8.1.1 Converting a DataFrame Back to JSON — `.to_json()`

```
df.to_json()
```

Result:

```
'{"employee_name":{"0":"James","1":"James"},"email":
{"0":"james@gmail.com","1":"james@gmail.com"},
"job_profile":{"0":{"title1":"Team Lead"},"1":{"title2":"Sr. Developer"}}}'
```

8.1.2 The `orient` Parameter in `.to_json()`

The `orient` parameter determines the **format and structure** of the resulting JSON string.

Since JSON is a flexible format, there are multiple ways to represent a table (rows and columns) in a nested string. Changing the `orient` changes how keys and values are paired up.

8.1.2.1 Common `orient` Options

Orient	Description	Best Use Case
`'records'`	A list of dictionaries: `[{column -> value}, ...]`	Web APIs — most frontend frameworks expect this
`'split'`	A dictionary containing index, columns, and data	Preserve exact structure without repeating columns
`'index'`	A dictionary of dictionaries: `{index -> {col:val}}`	Quick lookups by a specific ID or index
`'columns'`	A dictionary of dictionaries: `{col -> {idx:val}}`	Default — optimized for column-based access
`'values'`	Just the data in a nested list (no labels)	Minimal file size when labels aren't needed
`'table'`	Includes a JSON schema (data types and index info)	High-integrity data transfer between pandas

8.1.2.2 Code Examples

```
import pandas as pd

df = pd.DataFrame({
    'Name': ['Alice', 'Bob'],
    'Age':  [25, 30]
})

# 1. records — most common for web apps
print(df.to_json(orient='records'))
```

Result:

```
[{"Name":"Alice","Age":25},{"Name":"Bob","Age":30}]
```

```
# 2. split — separates metadata from data
print(df.to_json(orient='split'))
```

Result:

```
{"columns":["Name","Age"],"index":[0,1],"data":[["Alice",25],["Bob",30]]}
```

```
# 3. index — uses row index as the primary key
```

```
print(df.to_json(orient='index'))
```

Result:

```
{"0":{"Name":"Alice","Age":25},"1":{"Name":"Bob","Age":30}}
```

When to use which?

- Sending data to a **JavaScript app** → almost always use `orient='records'`
- **Saving to disk** to reopen in pandas later → use `orient='split'` or `'table'` (preserves index and column names more reliably)

8.1.2.3 Applying `orient` on Our DataFrame

```
df.to_json(orient="index")
```

Result:

```
'{"0":{"employee_name":"James","email":"james@gmail.com","job_profile":
{"title1":"Team Lead"}},
  "1":{"employee_name":"James","email":"james@gmail.com","job_profile":
{"title2":"Sr. Developer"}}}'
```

```
df.to_json(orient="records")
```

Result:

```
'[{"employee_name":"James","email":"james@gmail.com","job_profile":
{"title1":"Team Lead"}},
  {"employee_name":"James","email":"james@gmail.com","job_profile":
{"title2":"Sr. Developer"}}]'
```

8.2 Reading Data From a URL — `.read_csv(url)`

```
df = pd.read_csv(
    "https://archive.ics.uci.edu/ml/machine-learning-databases/wine/wine.data",
    header=None
```

```
)
df.head()
```

Result:

```
   0      1     2     3     4    5     6     7     8     9     10    11
12    13
0  1  14.23  1.71  2.43  15.6  127  2.80  3.06  0.28  2.29  5.64  1.04  3.92
1065
1  1  13.20  1.78  2.14  11.2  100  2.65  2.76  0.26  1.28  4.38  1.05  3.40
1050
2  1  13.16  2.36  2.67  18.6  101  2.80  3.24  0.30  2.81  5.68  1.03  3.17
1185
3  1  14.37  1.95  2.50  16.8  113  3.85  3.49  0.24  2.18  7.80  0.86  3.45
1480
4  1  13.24  2.59  2.87  21.0  118  2.80  2.69  0.39  1.82  4.32  1.04  2.93
735
```

Note: `header=None` is used when the data file does **not** contain column names in the first row. Without it, pandas will automatically treat the first row of data as the column headers.

8.2.1 Converting Back to CSV

```
df.to_csv("wine.csv")
# New file wine.csv is created and stored with the df data
```

8.3 Reading Data From HTML — `.read_html()`

```
url = "https://www.fdic.gov/bank/individual/failed/banklist.html"
df = pd.read_html(url)

df[0]    # df[0] gives you the first table extracted from the website
```

Result:

```
                              Bank Name       City      State   Cert  ...
Closing Date   Fund
0   Metropolitan Capital Bank & Trust    Chicago   Illinois  57488  ...
January 30, 2026  10550
1       The Santa Anna National Bank  Santa Anna      Texas   5520  ...
June 27, 2025  10549
2              Pulaski Savings Bank    Chicago   Illinois  28611  ...
January 17, 2025  10548
```

```
...
24    Seaway Bank and Trust Company    Chicago   Illinois  19328  ...
January 27, 2017  10524
```

8.3.1 Another HTML Example — Wikipedia Table with Filters

```
url = "https://en.wikipedia.org/wiki/Mobile_country_code"
df = pd.read_html(
    url,
    match="Country",        # Only extract tables that contain the word "Country"
    header=0,               # First row is the header
    storage_options={"User-Agent": "Mozilla/5.0"}
)
df[0]
```

Result:

```
   Mobile country code          Country ISO 3166  ...  Remarks
0                  289      A Abkhazia   GE-AB    ...  MCC is not listed by
ITU
1                  412      Afghanistan      AF   ...  NaN
2                  276          Albania      AL   ...  NaN
...
251                648         Zimbabwe      ZW   ...  NaN
252 rows × 6 columns
```

8.4 Reading Data From Excel — `.read_excel()`

```
df_excel = pd.read_excel('Data_excel.xlsx')
df_excel
```

Result:

```
      Name  Age
0    Sher   19
1  Hassan   20
2    Khan   21
```

8.5 Pickle — Serializing Python Objects

Pickle is primarily used for **serializing and deserializing** a Python object structure. In other words, it converts a Python object into a byte stream to:

- Store it in a file / database
- Maintain program state across sessions
- Transport data over the network

8.5.1 Converting a DataFrame to a Pickle File

```
df_excel.to_pickle('df_pickle')
# File saved with the name: df_pickle
```

> **Note:** There are many more methods available under `.read_xxx()` and `.to_xxx()` in pandas for various file formats.

---

9. Data Visualization With Matplotlib

> **"Matplotlib is a powerful plotting library for Python that enables the creation of static, animated, and interactive visualizations. It is widely used for data visualization in data science and machine learning."**

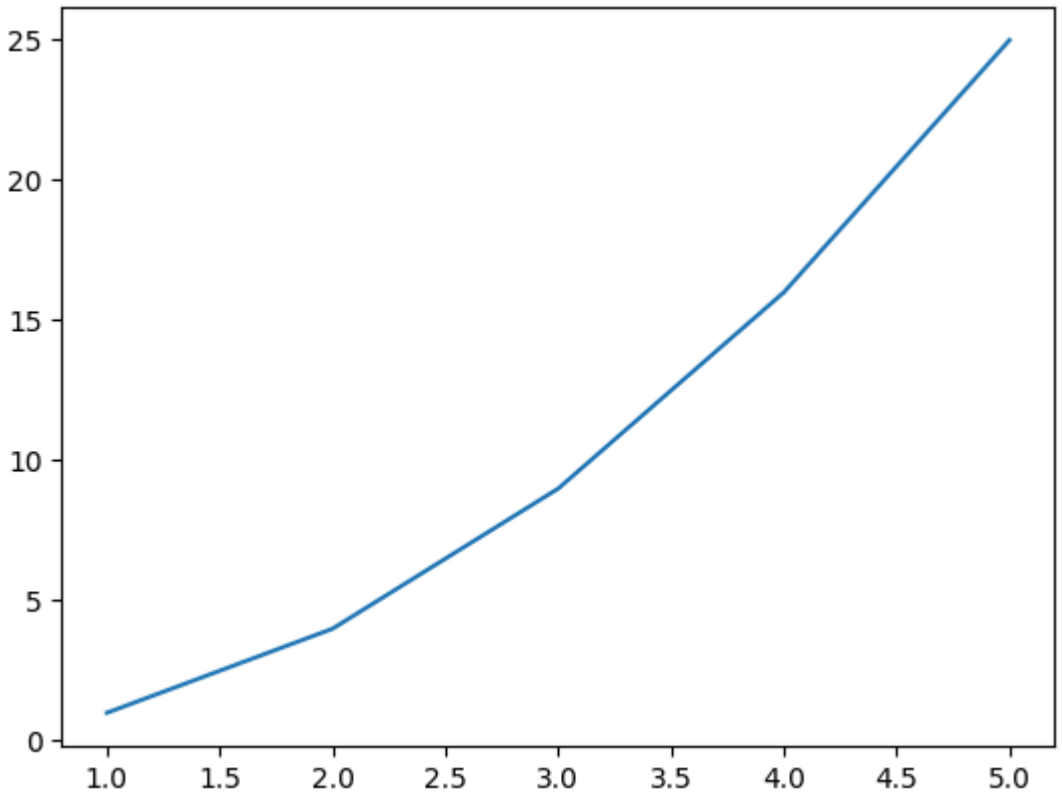
9.1 Importing Matplotlib

```
import matplotlib.pyplot as plt

# Note: Most of the data visualization functions are available in pyplot
```

9.2 Line Plot

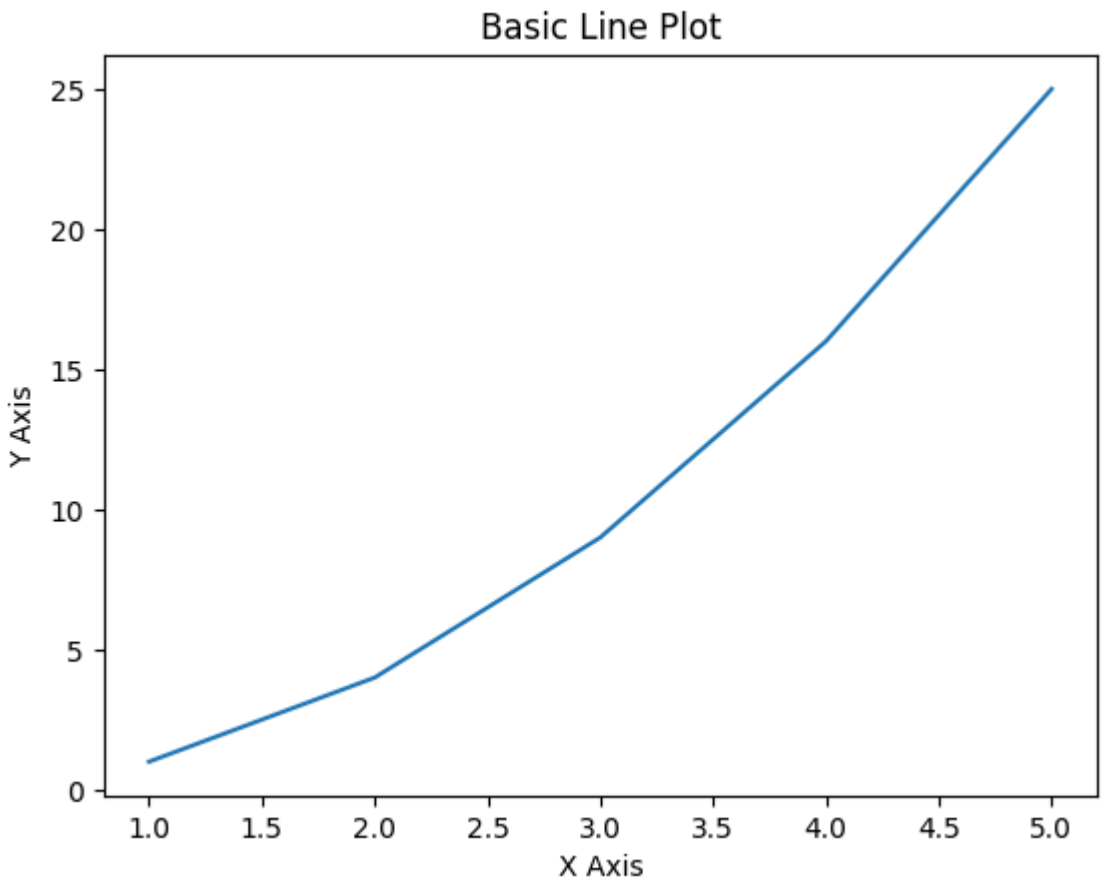
Basic Line Plot

```
x = [1, 2, 3, 4, 5]  # X axis
y = [1, 4, 9, 16, 25]  # Y axis

# Creating a line plot
plt.plot(x, y)
plt.show()
```



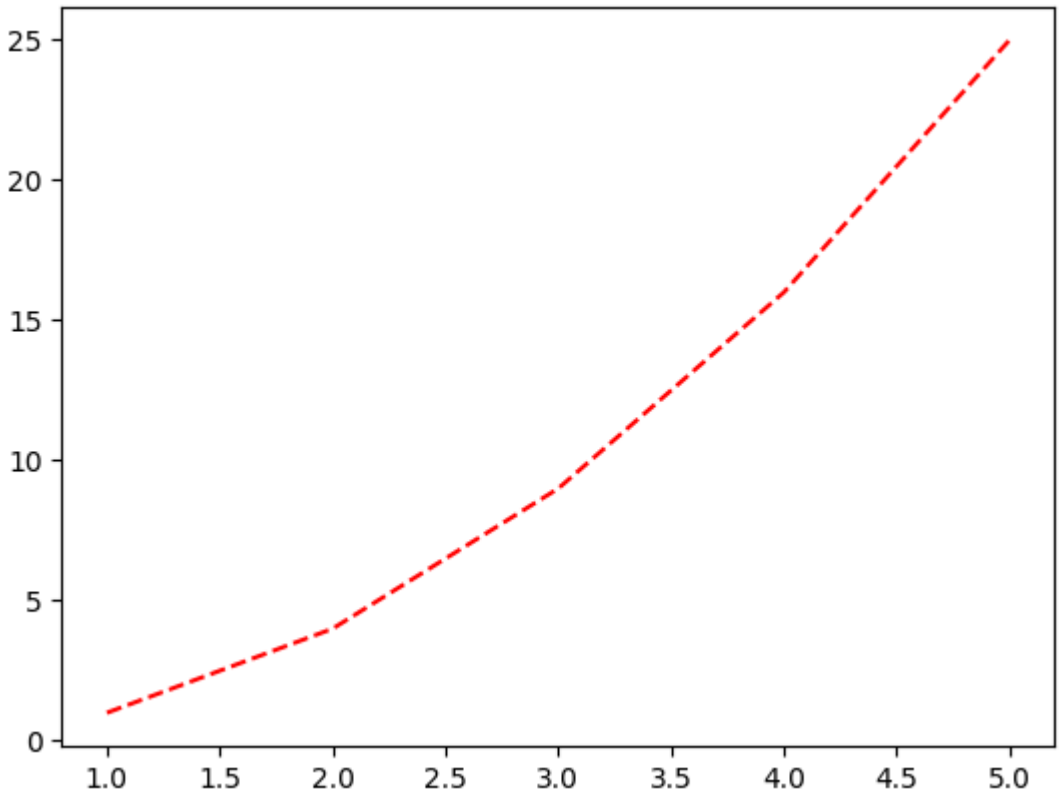
Line Plot With Labels and Title

```
x = [1, 2, 3, 4, 5]  # X axis
y = [1, 4, 9, 16, 25]  # Y axis

plt.plot(x, y)
plt.xlabel('X Axis')        # X axis name
plt.ylabel('Y Axis')        # Y axis name
plt.title("Basic Line Plot") # Name of the graph
plt.show()                   # Show the graph
```



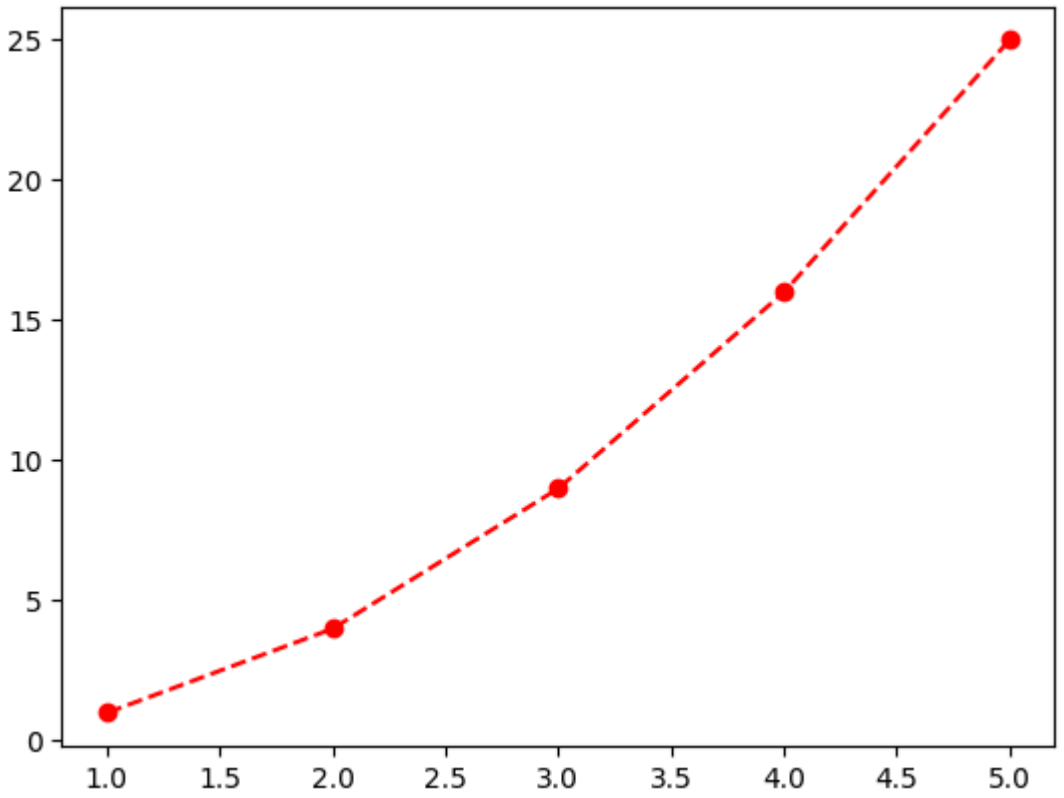
Customized Line Plot — Color and Linestyle

```
x = [1, 2, 3, 4, 5]
y = [1, 4, 9, 16, 25]

plt.plot(x, y, color='red', linestyle='--')

# Supported values for linestyle:
# '-', '--', '-.', ':', 'None', ' ', '',
# 'solid', 'dashed', 'dashdot', 'dotted'
```



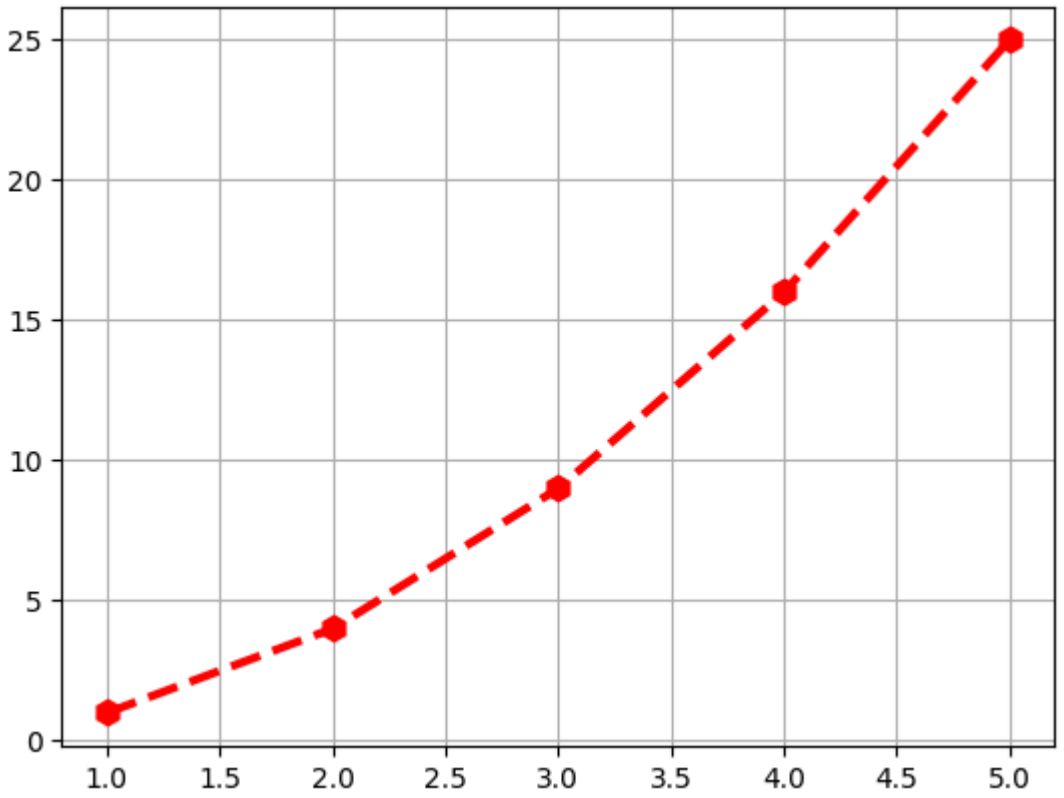
Adding Markers

```
# Using the marker parameter
plt.plot(x, y, color='red', linestyle='--', marker='o')
# Puts 'o' on the values of x and y in the graph
# Other marker options: *, s, m, h, etc.
```



Line Width, Marker Size, and Grid

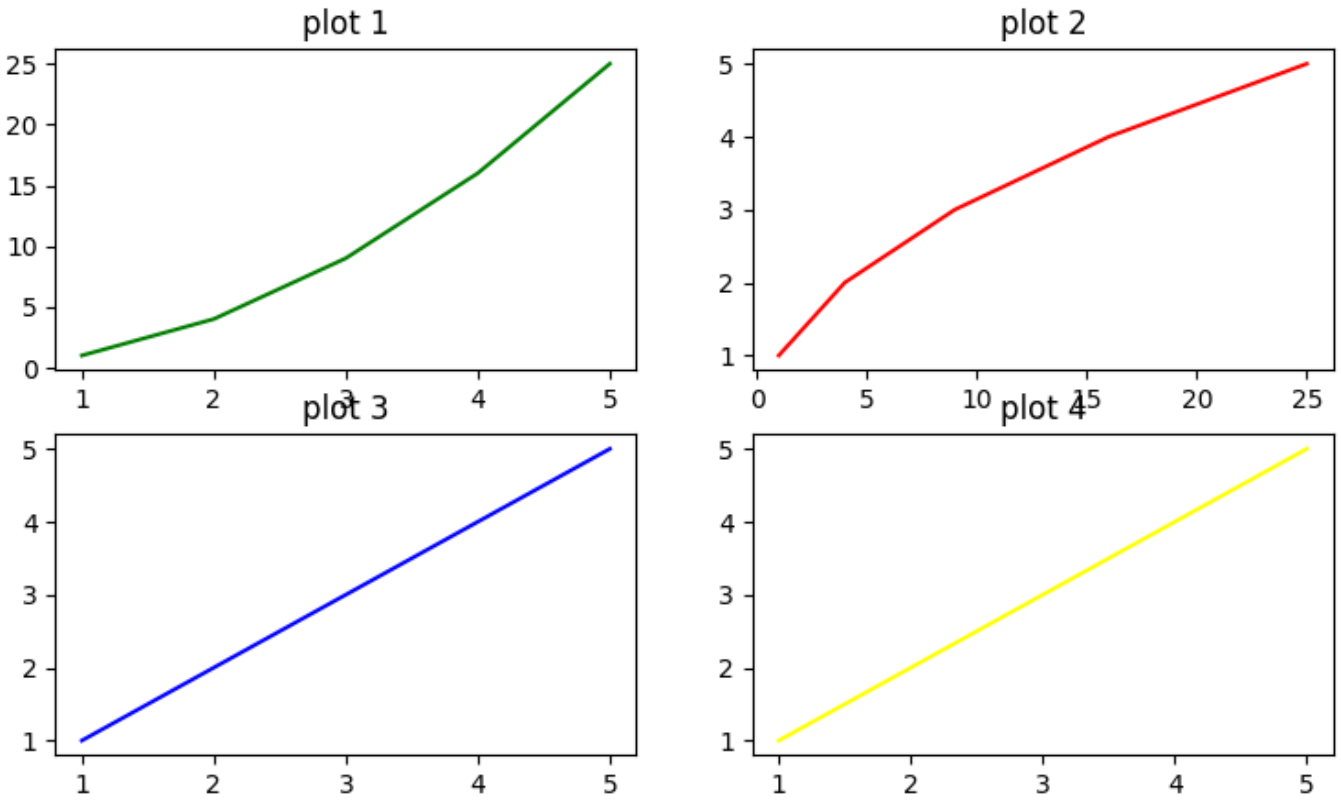
```
# line width and marker size parameters
plt.plot(x, y, color='red', linestyle='--', marker='h', linewidth=3, markersize=9)

# We can also add a grid
plt.grid(True)
```



9.3 Multiple Plots (Subplots)

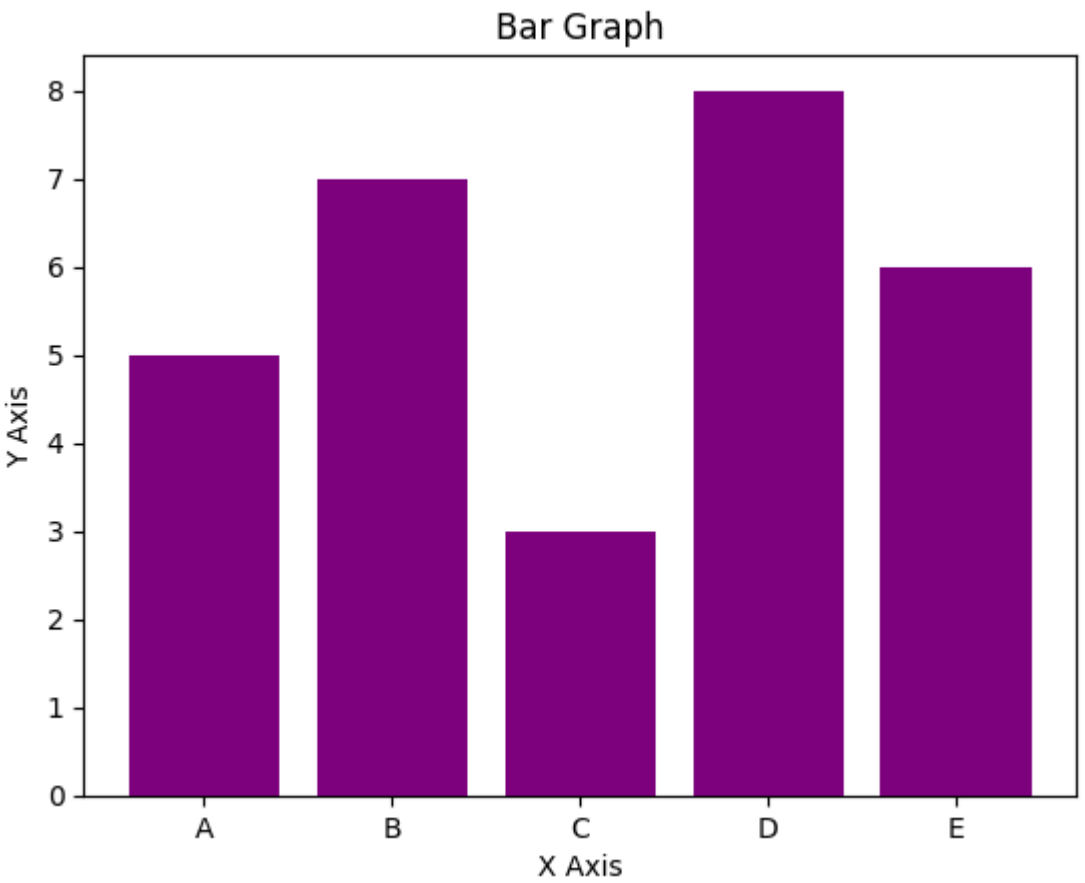
```
# Sample data
x  = [1, 2, 3, 4, 5]
y1 = [1, 4, 9, 16, 25]
y2 = [1, 2, 3, 4, 5]

# Creating Parent Figure
plt.figure(figsize=(9, 5))
# (9,5) is the size of the figure in inches
# This is the main parent figure that includes independent subplots

# Plotting subplots
# plt.subplot(rows, cols, position)
# - First two args (2,2): how many rows and columns of subplots to create
# - Third arg: the position of this subplot in the parent figure

plt.subplot(2, 2, 1)
plt.plot(x, y1, color='green')
plt.title("plot 1")

plt.subplot(2, 2, 2)
plt.plot(y1, x, color='red')
plt.title("plot 2")

plt.subplot(2, 2, 3)
plt.plot(x, y2, color='blue')
plt.title("plot 3")
```

```
plt.subplot(2, 2, 4)
plt.plot(x, y2, color='yellow')
plt.title("plot 4")
```



9.4 Bar Plot

```
categories = ['A', 'B', 'C', 'D', 'E']
values = [5, 7, 3, 8, 6]

# Creating a Bar Plot
plt.bar(categories, values, color='purple')

# Additional informatory parameters
plt.xlabel('X Axis')    # X axis name
plt.ylabel('Y Axis')    # Y axis name
plt.title("Bar Graph")  # Name of the graph
plt.show()
```

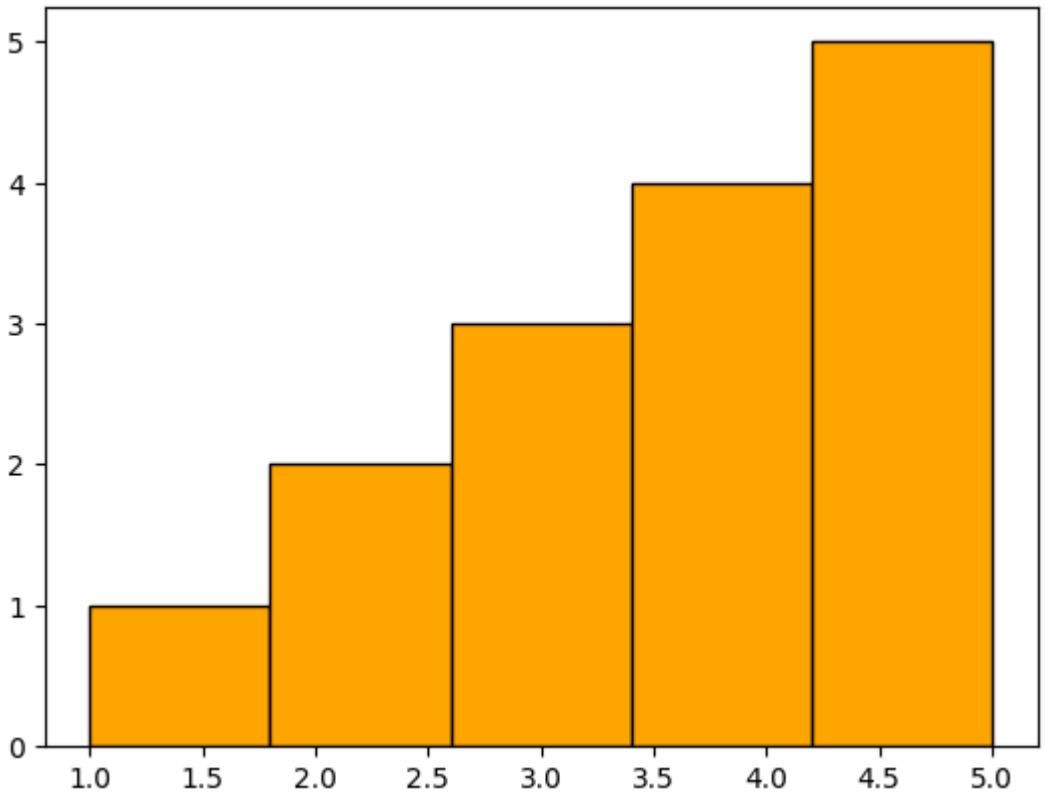


9.5 Histograms

> **"Histograms are used to represent the distribution of a dataset. They divide the data into bins and count the number of data points in each bin."**

```
# Sample data
data = [1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5]

# Creating a histogram
# bins=5 means: take this data and divide it into 5 bins
plt.hist(data, bins=5, color='orange', edgecolor="black")
```

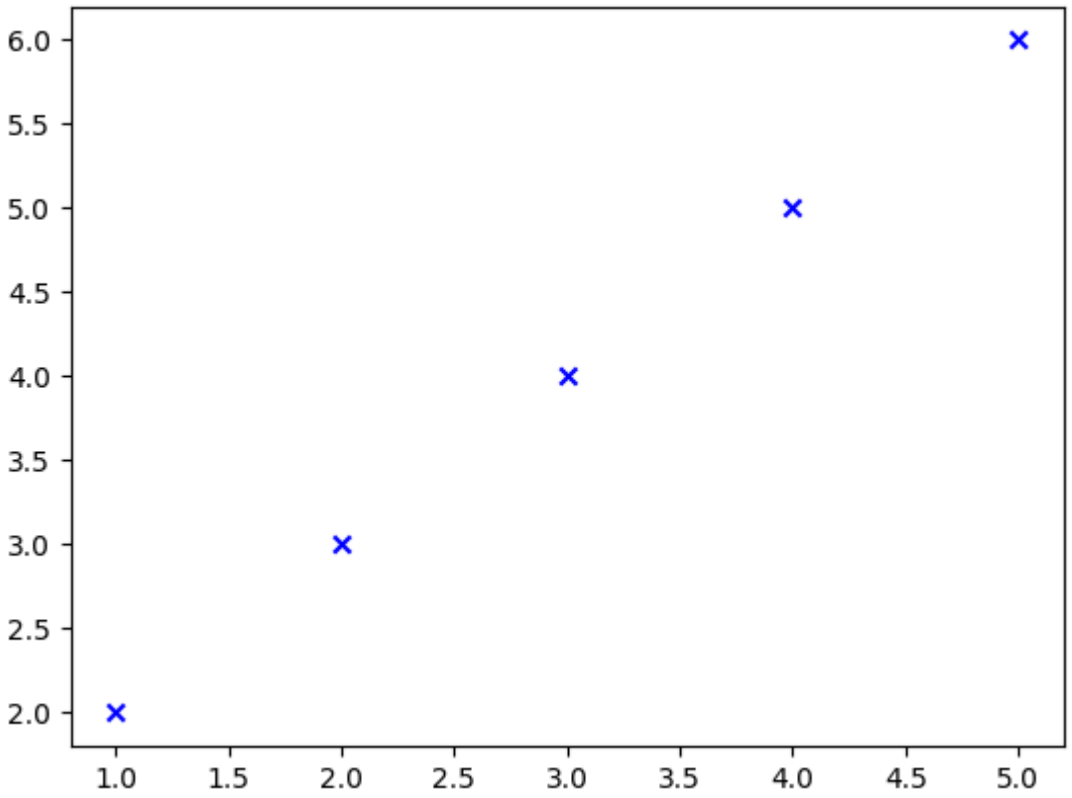
> **Result:**
>
> ```
> (array([1., 2., 3., 4., 5.]),
> array([1. , 1.8, 2.6, 3.4, 4.2, 5. ]),
> <BarContainer object of 5 artists>)
> ```



9.6 Scatter Plot

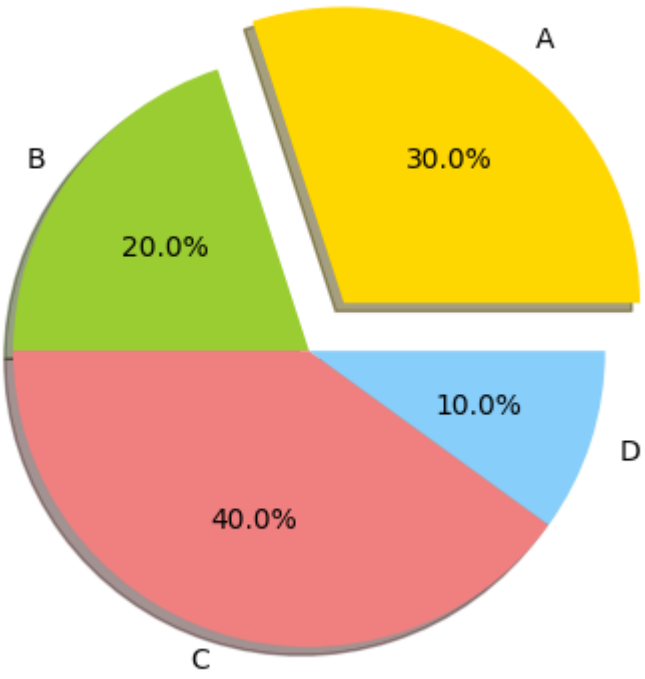
> **"A scatter plot is a type of data visualization that uses dots to represent the values for two different variables. It is the go-to tool for identifying relationships or correlations between variables."**

```
# Sample Data
x = [1, 2, 3, 4, 5]
y = [2, 3, 4, 5, 6]

plt.scatter(x, y, color="blue", marker="x")
```



9.7 Pie Chart

```
labels  = ['A', 'B', 'C', 'D']
sizes   = [30, 20, 40, 10]
colors  = ['gold', 'yellowgreen', 'lightcoral', 'lightskyblue']
explode = (0.2, 0, 0, 0)  # Moves the first slice outward for decoration

plt.pie(
    sizes,
    explode=explode,
    labels=labels,
    colors=colors,
    autopct="%1.1f%%",   # Format: XX.X% (one decimal place)
    shadow=True
)
```



9.8 Practical Example — Sales Data Visualization

```
import pandas as pd

sales_data_df = pd.read_csv('sales_data.csv')
sales_data_df.head()
```

9.8.1 Total Sales by Item Type (Bar Chart)

Let's group based on Item Type, sum the Total Revenue (which gives total sales), and then make a bar graph.

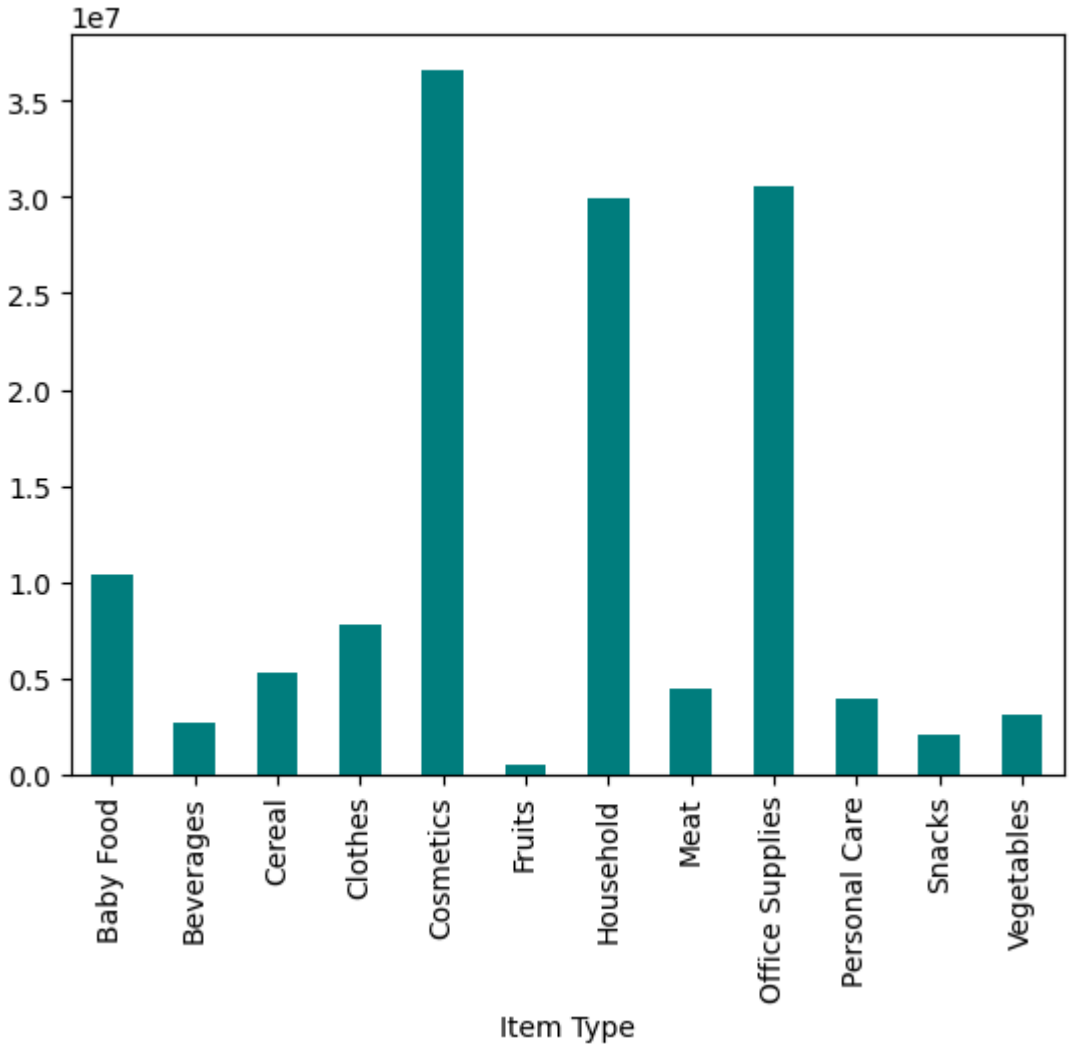
```
## Plot total sales by Item Type
sales_data_df.info()

total_sales_by_products = sales_data_df.groupby('Item Type')['Total
Revenue'].sum()
print(total_sales_by_products)
```

Result (`sales_data_df.info()`):

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 14 columns):
 #   Column          Non-Null Count  Dtype
```

```
---  ------          --------------  -----
 0   Region          100 non-null    str
 1   Country         100 non-null    str
 2   Item Type       100 non-null    str
 3   Sales Channel   100 non-null    str
 4   Order Priority  100 non-null    str
 5   Order Date      100 non-null    str
 6   Order ID        100 non-null    int64
 7   Ship Date       100 non-null    str
 8   Units Sold      100 non-null    int64
 9   Unit Price       96 non-null    float64
 10  Unit Cost       100 non-null    float64
 11  Total Revenue   100 non-null    float64
 12  Total Cost      100 non-null    float64
 13  Total Profit    100 non-null    float64
dtypes: float64(5), int64(2), str(7)
memory usage: 11.1 KB
```

```python
# Creating the bar graph
total_sales_by_products.plot(kind='bar', color='teal')
```



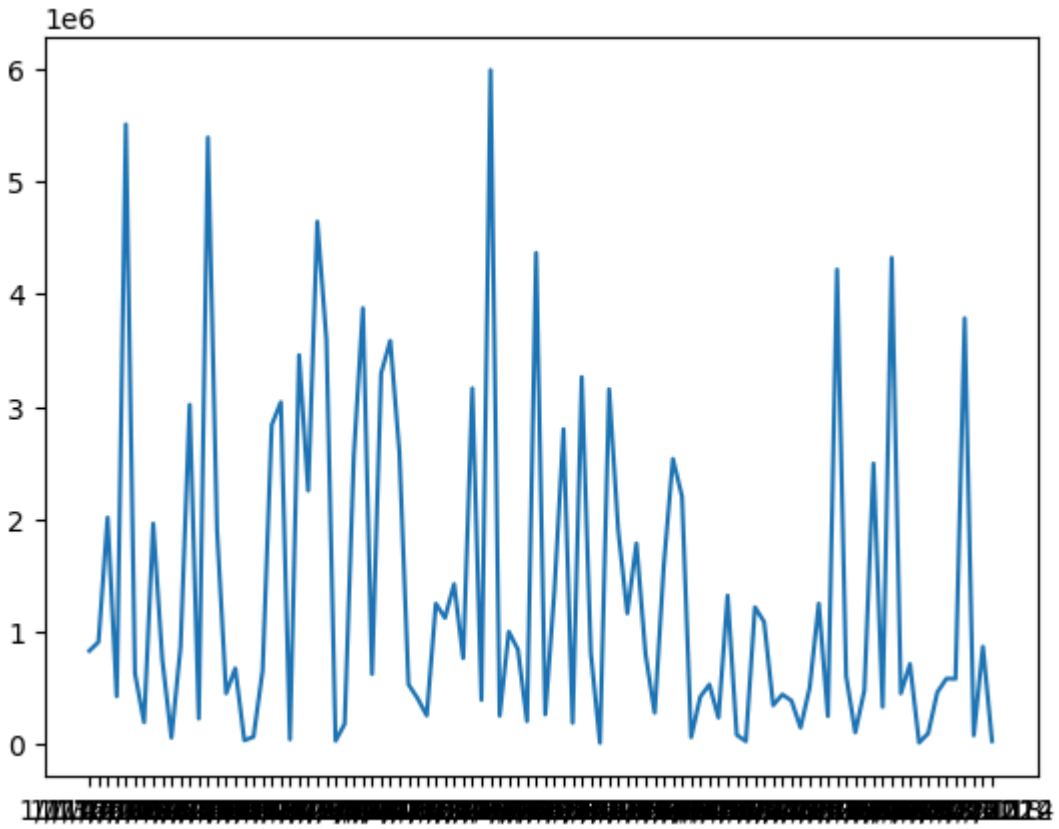
9.8.2 Sales Trend Over Time (Line Chart)

```
# Plot sales trend over time

sales_trend = sales_data_df.groupby('Order Date')['Total
Revenue'].sum().reset_index()
# reset_index() is used because after groupby, "Order Date" becomes the Index, not
a column.
# We reset it so we can use it as the X axis directly.

sales_trend
```

Result (first few rows):

```
   Order Date    Total Revenue
0  1/11/2012      824431.86
1  1/13/2017      902980.64
2  1/14/2017     2011149.63
...
```

```
plt.plot(sales_trend['Order Date'], sales_trend['Total Revenue'])
```



10. Data Visualization With Seaborn

> **"Seaborn is a Python visualization library based on Matplotlib that provides a high-level interface for drawing attractive and informative statistical graphics. Seaborn helps in creating complex visualizations with just a few lines of code."**

10.1 Importing Seaborn

```
import seaborn as sns
```

10.2 Built-in Datasets

Seaborn comes with a collection of **built-in sample datasets** that are used for practicing data visualization and testing library features.

```
sns.get_dataset_names()
```

> **Result:**
>
> ```
> ['anagrams', 'anscombe', 'attention', 'brain_networks', 'car_crashes',
> 'diamonds', 'dots', 'dowjones', 'exercise', 'flights', 'fmri',
> 'geyser', 'glue', 'healthexp', 'iris', 'mpg', 'penguins', 'planets',
> 'seaice', 'taxis', 'tips', 'titanic']
> ```

Loading a Dataset

```
tips = sns.load_dataset('tips')
print(type(tips))
tips
```

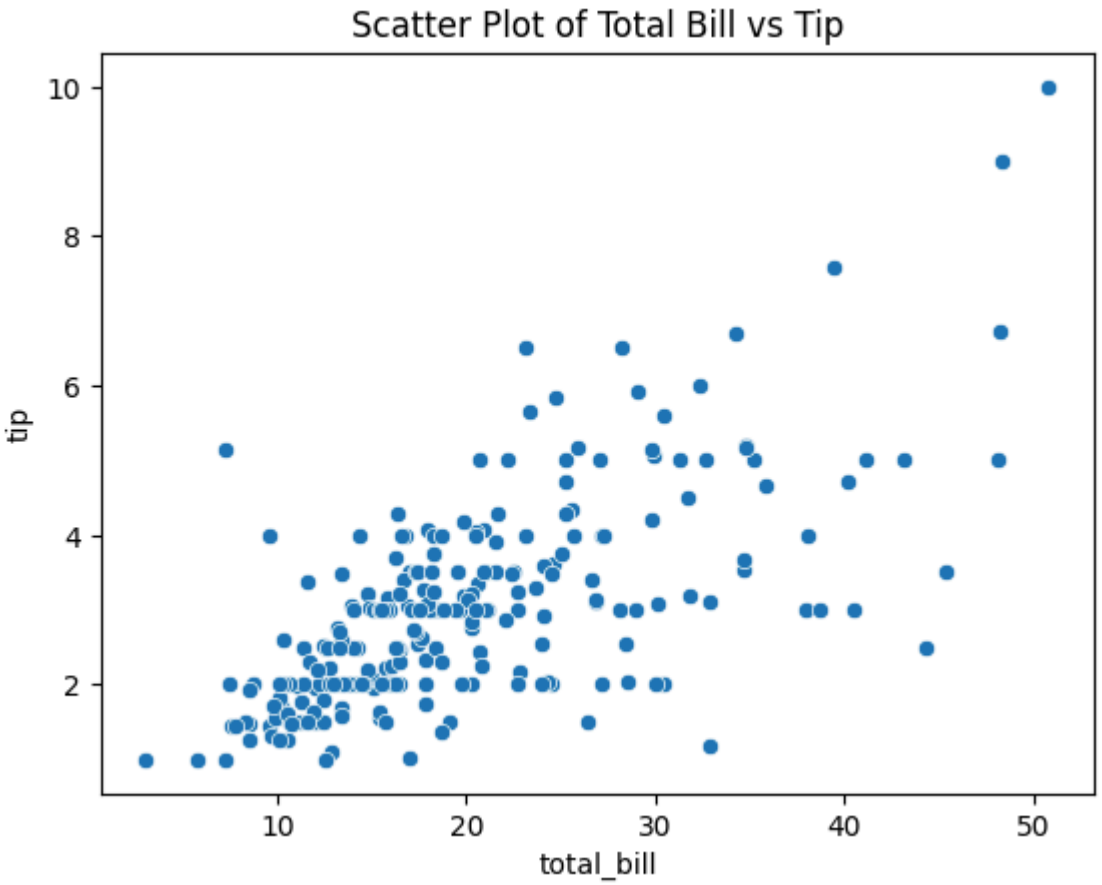
> **Result:**
>
> ```
> <class 'pandas.DataFrame'>
> ```

10.3 Common Syntax for Seaborn Plots

```
seaborn.plotname(x='Xname', y='Yname', data=DataFrameName)
```

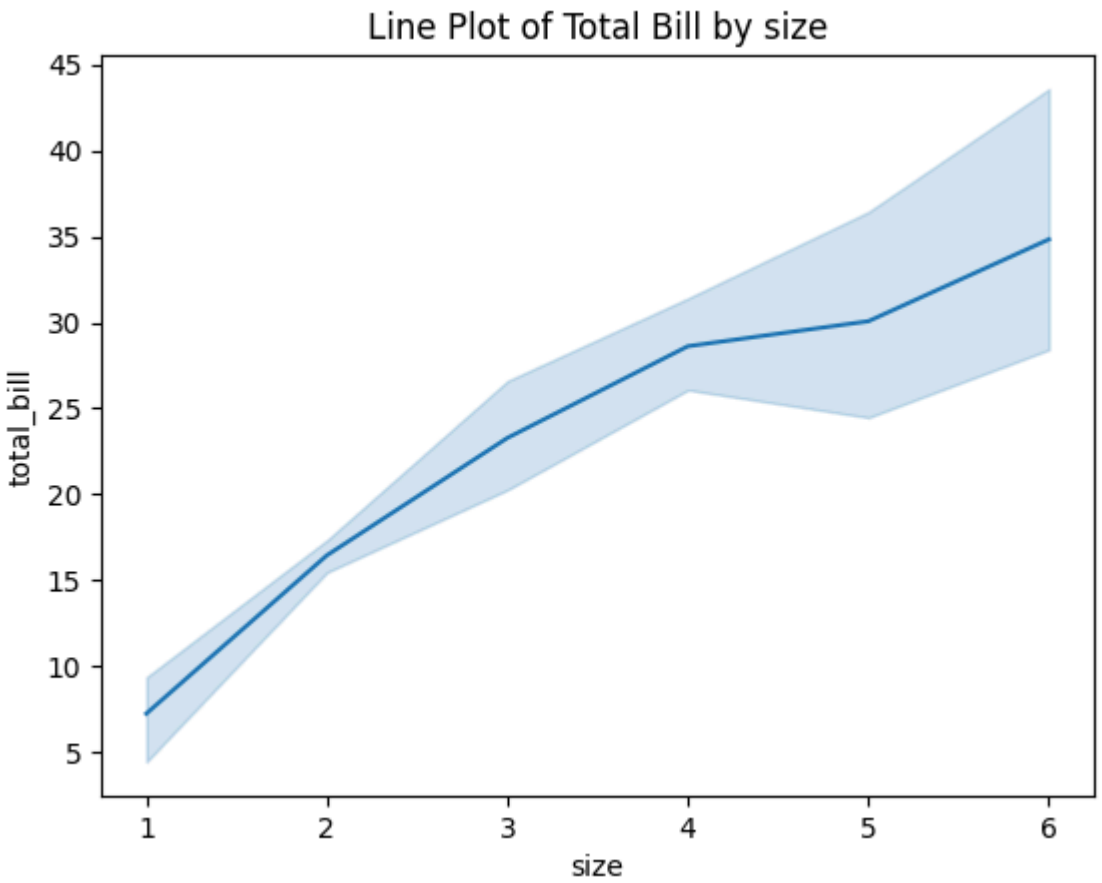
10.4 Scatter Plot

```
import matplotlib.pyplot as plt

# sns.scatterplot(x='xname', y='yname', data=dataframename)
sns.scatterplot(x='total_bill', y='tip', data=tips)
plt.title("Scatter Plot of Total Bill vs Tip")
plt.show()
```



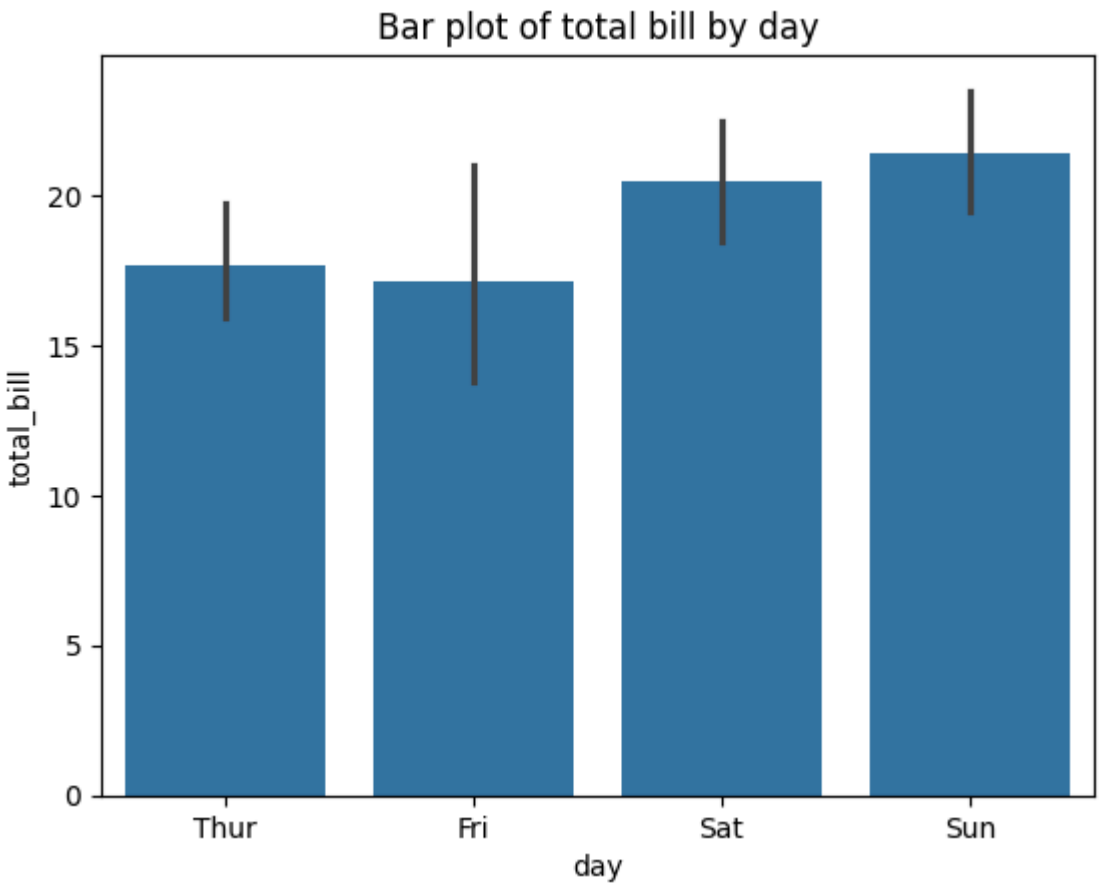
10.5 Line Plot

```
# sns.lineplot(x='xname', y='yname', data=dataframename)
sns.lineplot(x='size', y='total_bill', data=tips)
plt.title("Line Plot of Total Bill by size")
plt.show()
```



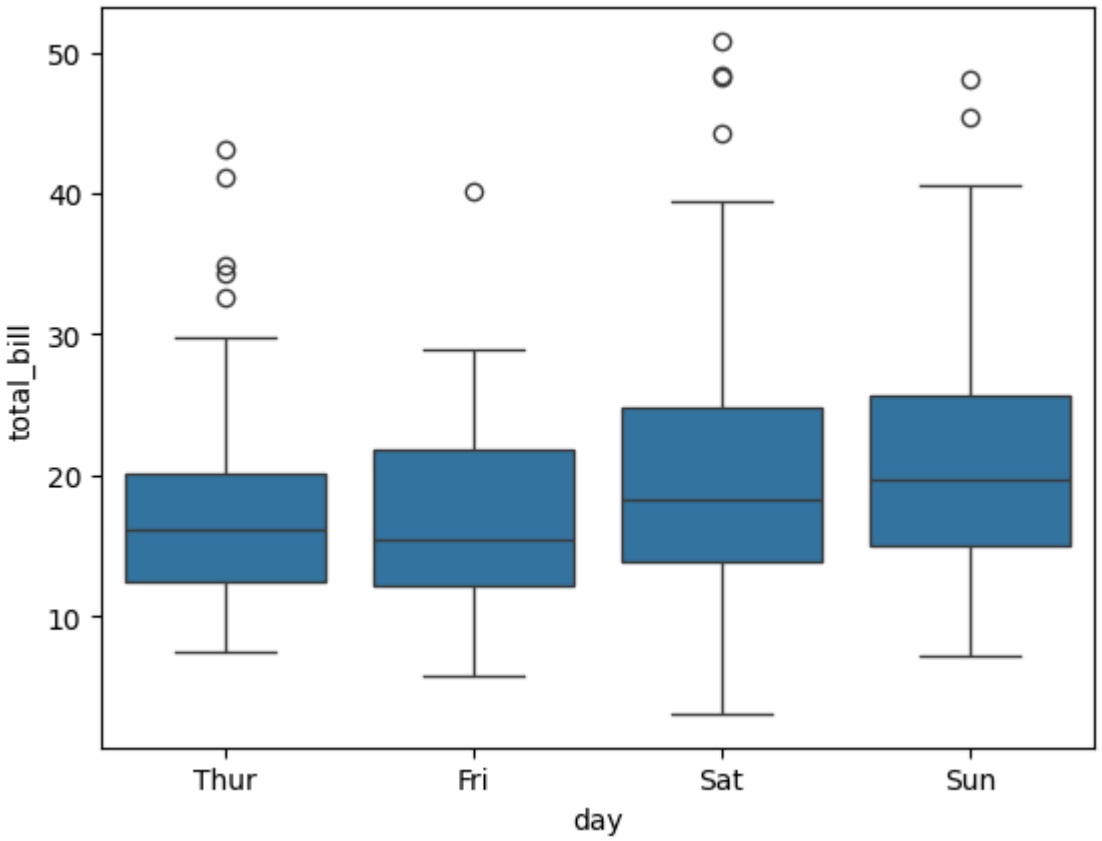
10.6 Bar Plot (Categorical Plot)

```
sns.barplot(x='day', y='total_bill', data=tips)
plt.title("Bar plot of total bill by day")
plt.show()
```



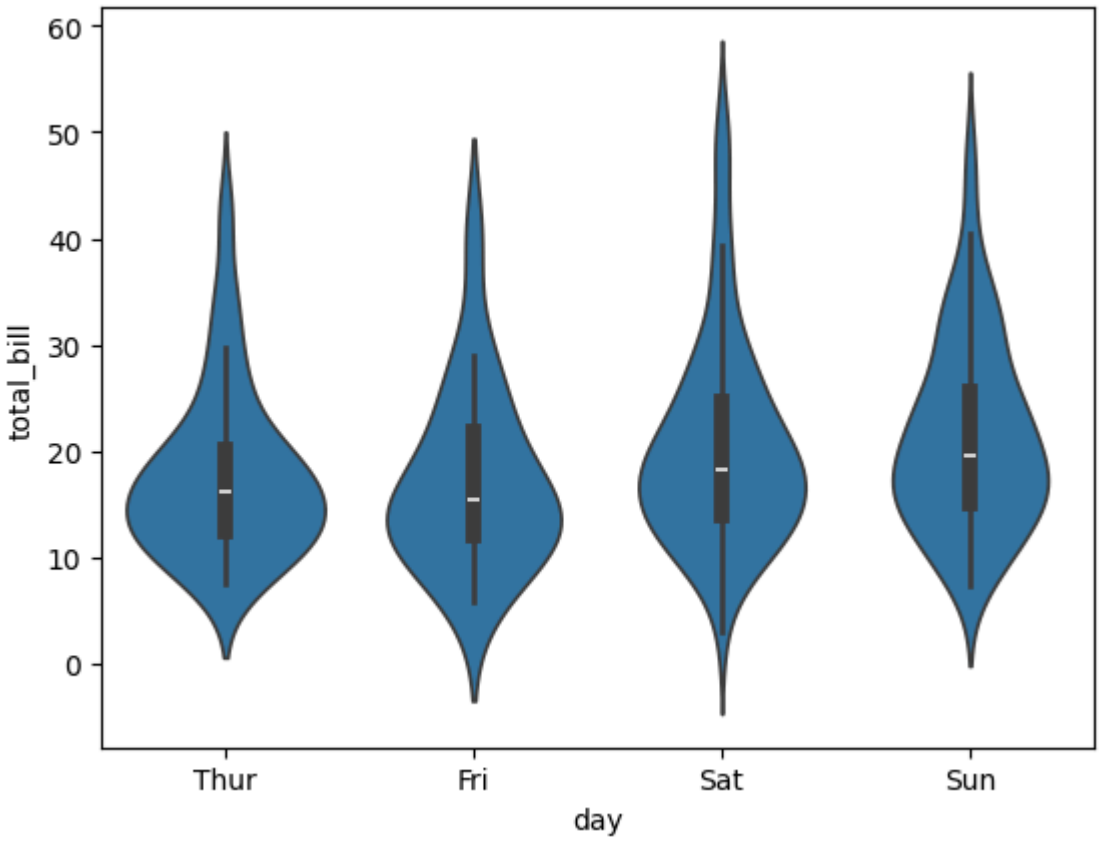
10.7 Box Plot

```
sns.boxplot(x='day', y='total_bill', data=tips)
plt.show()
```



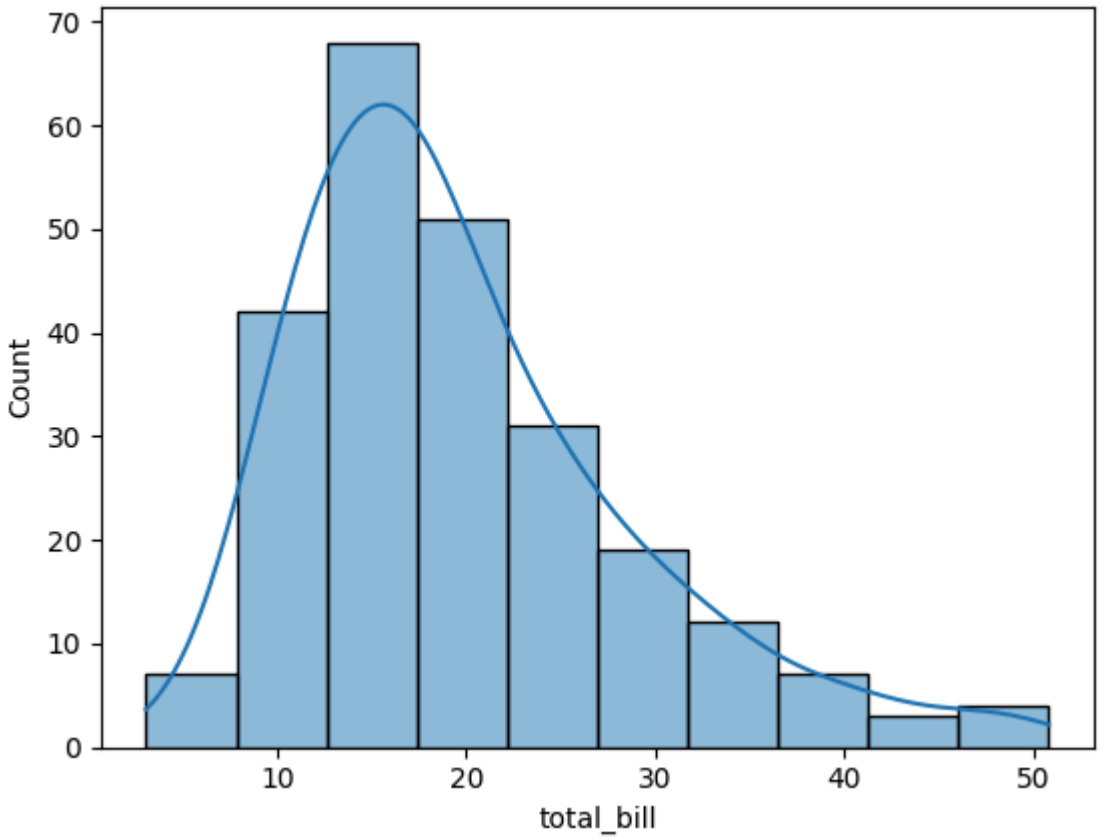
10.8 Violin Plot

```
sns.violinplot(x='day', y='total_bill', data=tips)
plt.show()
```



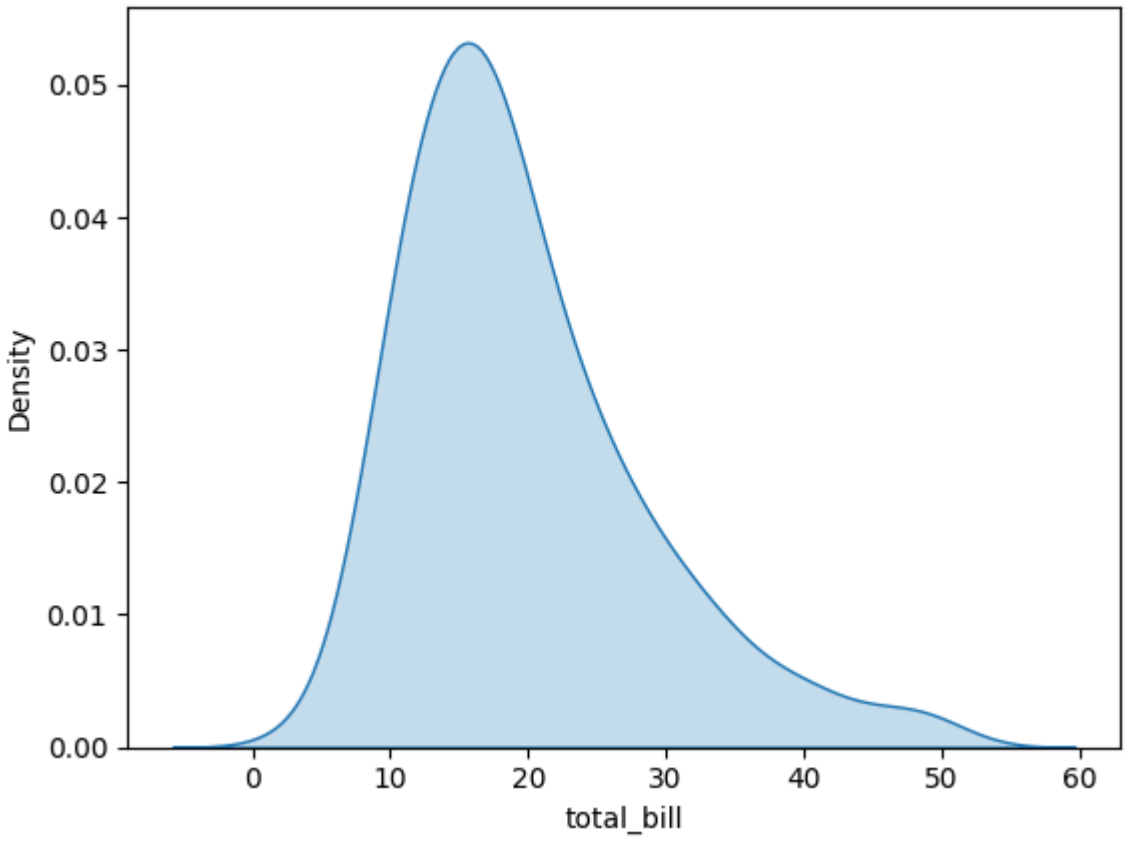
10.9 Histogram

```
sns.histplot(tips['total_bill'], bins=10, kde=True)
# kde=True adds the smooth curved line on top of the histogram
plt.show()
```



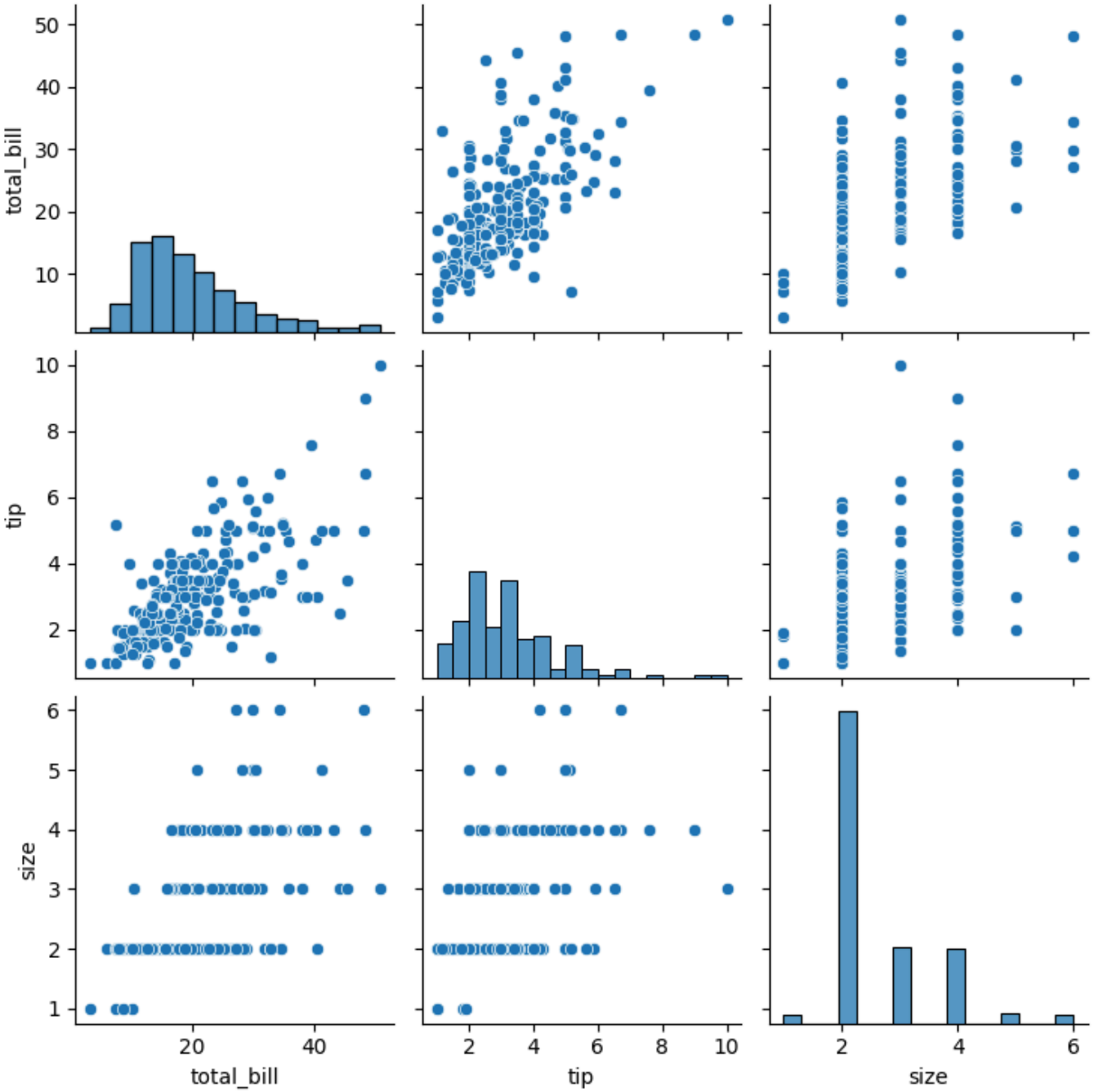
10.10 KDE Plot (Kernel Density Estimate)

```
sns.kdeplot(tips['total_bill'], fill=True)
plt.show()
```



10.11 Pair Plot

```
# Shows the relationship of all numerical columns with each other
sns.pairplot(tips)
plt.show()
```



10.12 Heatmap

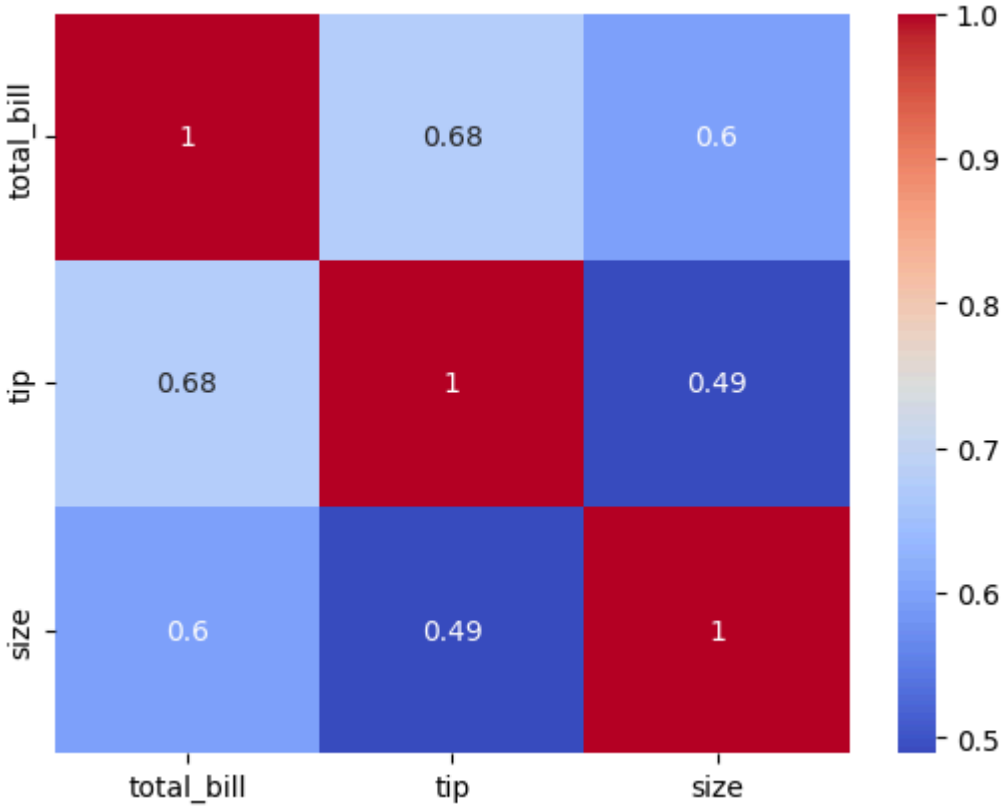
A heatmap is used to visualize the **correlation** between numerical columns.

Step 1 — Compute the Correlation Matrix

```
corr = tips[['total_bill', 'tip', 'size']].corr()
# .corr() computes the pairwise correlation of numerical columns
# Only works on numerical data
corr
```

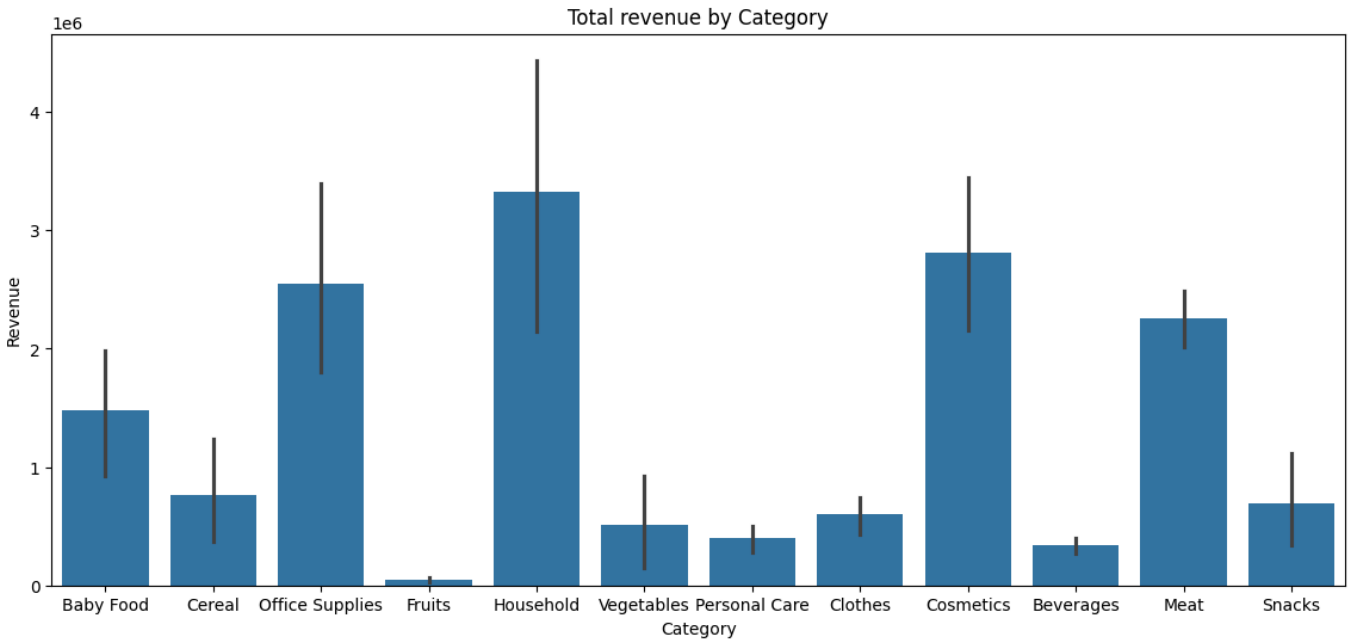
Step 2 — Plot the Heatmap

```
sns.heatmap(corr, annot=True, cmap='coolwarm')
# annot=True displays the correlation values inside each cell
# cmap='coolwarm' sets the color scheme
```



10.13 Practical Example — Sales Data Visualization

```
import pandas as pd

myData = pd.read_csv('sales_data.csv')
myData.head()

# Using a bar graph to show Total Revenue by Item Type

plt.figure(figsize=(14, 6))                              # First create the
figure
sns.barplot(x='Item Type', y='Total Revenue', data=myData)  # Then plot the
subplot
plt.title('Total revenue by Category')                   # Then add title
plt.xlabel('Category')                                   # Then label X axis
plt.ylabel('Revenue')                                    # Then label Y axis
plt.show()                                               # Then show it
```


Total revenue by Category
1e6
4
3
2
1
0
Revenue
Baby Food
Cereal
Office Supplies
Fruits
Household
Vegetables
Personal Care
Clothes
Cosmetics
Beverages
Meat
Snacks
Category

---

11. SQL and SQLite With Python

> **"SQL (Structured Query Language) is a standard language for managing and manipulating relational databases. SQLite is a self-contained, serverless, and zero-configuration database engine that is widely used for embedded database systems."**

11.1 Importing SQLite3

```
import sqlite3
```

11.2 Connecting to a Database

```
connection = sqlite3.connect('example.db')
connection
```

> **Result:**
>
> ```
> <sqlite3.Connection at 0x22023994c70>
> ```

> **Note:** If `example.db` does not exist, SQLite will automatically create it.

11.3 Creating a Cursor

A cursor acts as a control structure or **"pointer"** that allows you to execute SQL queries and fetch the resulting rows one by one from the database.

```
cursor = connection.cursor()
```

11.4 Creating a Table

```
cursor.execute('''
    CREATE TABLE IF NOT EXISTS employees (
        id         INTEGER PRIMARY KEY,
        name       TEXT    NOT NULL,
        age        INTEGER,
        department TEXT
```

```
    )
''')

# Commit the changes
connection.commit()
```

> **Note:** `IF NOT EXISTS` ensures the table is only created if it doesn't already exist — prevents errors on re-runs.

11.5 General Query Syntax

Any SQL query is executed by passing it as a string to `cursor.execute()`.

```
cursor.execute('''
    (SQL Query here)
''')
```

11.6 Inserting Data

Inserting Single Rows

```
cursor.execute('''
    INSERT INTO employees (name, age, department)
    VALUES ('Sher', 20, 'Data Scientist')
''')

cursor.execute('''
    INSERT INTO employees (name, age, department)
    VALUES ('John', 25, 'Data Engineer')
''')

cursor.execute('''
    INSERT INTO employees (name, age, department)
    VALUES ('Jane', 30, 'Machine Learning')
''')

# Commit the changes
connection.commit()
```

11.7 Querying Data — `SELECT`

```
cursor.execute('SELECT * FROM employees')
rows = cursor.fetchall()    # Get all the records (rows)
```

```
for row in rows:
    print(row)
```

Result:

```
(1, 'Sher', 20, 'Data Scientist')
(2, 'John', 25, 'Data Engineer')
(3, 'Jane', 30, 'Machine Learning')
```

11.8 Updating Data — UPDATE

```
cursor.execute('''
    UPDATE employees
    SET age = 27
    WHERE name = "Sher"
''')

connection.commit()
```

Verifying the Update

```
cursor.execute('SELECT * FROM employees')
rows = cursor.fetchall()

for row in rows:
    print(row)
```

Result:

```
(1, 'Sher', 27, 'Data Scientist')
(2, 'John', 25, 'Data Engineer')
(3, 'Jane', 30, 'Machine Learning')
```

11.9 Deleting Data — DELETE

```
cursor.execute('''
    DELETE FROM employees
    WHERE name = "John"
''')
```

```
connection.commit()
```

Verifying the Deletion

```
cursor.execute('SELECT * FROM employees')
rows = cursor.fetchall()

for row in rows:
    print(row)
```

> **Result:**
>
> ```
> (1, 'Sher', 27, 'Data Scientist')
> (3, 'Jane', 30, 'Machine Learning')
> ```

11.10 Inserting Bulk Data — `executemany()`

Instead of calling `execute()` repeatedly, use `executemany()` to insert multiple rows at once. The `?` placeholders are filled in from each tuple in the list.

```
employees_data = [
    (1,  'Sher',    27, 'Data Scientist'),
    (2,  'jade',    29, 'Soft. Engineer'),
    (3,  'Jane',    30, 'Machine Learning'),
    (4,  'Alice',   24, 'Data Analyst'),
    (5,  'Bob',     29, 'Data Engineer'),
    (6,  'Charlie', 31, 'Machine Learning Engineer'),
    (7,  'Diana',   26, 'Business Analyst'),
    (8,  'Ethan',   35, 'DevOps Engineer'),
    (9,  'Fiona',   28, 'Software Engineer'),
    (10, 'George',  33, 'Product Manager'),
    (11, 'Hannah',  30, 'QA Engineer'),
    (12, 'Ivan',    27, 'Database Administrator')
]

cursor.executemany('''
    INSERT INTO employees (id, name, age, department)
    VALUES (?, ?, ?, ?)
''', employees_data)

connection.commit()
```

Verifying the Bulk Insert

```python
cursor.execute('SELECT * FROM employees')
rows = cursor.fetchall()

for row in rows:
    print(row)
```

Result:

```
(1,  'Sher',    27, 'Data Scientist')
(2,  'jade',    29, 'Soft. Engineer')
(3,  'Jane',    30, 'Machine Learning')
(4,  'Alice',   24, 'Data Analyst')
(5,  'Bob',     29, 'Data Engineer')
(6,  'Charlie', 31, 'Machine Learning Engineer')
(7,  'Diana',   26, 'Business Analyst')
(8,  'Ethan',   35, 'DevOps Engineer')
(9,  'Fiona',   28, 'Software Engineer')
(10, 'George',  33, 'Product Manager')
(11, 'Hannah',  30, 'QA Engineer')
(12, 'Ivan',    27, 'Database Administrator')
```

11.11 Closing the Connection

Always close the connection when you are done to free up resources.

```python
connection.close()
```

Quick Reference — SQLite3 Workflow

Step	Code	Purpose
Connect	`sqlite3.connect('file.db')`	Open / create a database
Cursor	`connection.cursor()`	Create a query pointer
Create Table	`cursor.execute('CREATE TABLE ...')`	Define table structure
Insert One	`cursor.execute('INSERT INTO ...')`	Add a single row
Insert Many	`cursor.executemany('INSERT INTO ...', data)`	Add multiple rows at once
Query	`cursor.execute('SELECT * FROM ...')`	Run a SELECT query
Fetch	`cursor.fetchall()`	Retrieve all result rows
Update	`cursor.execute('UPDATE ... SET ...')`	Modify existing rows

Step	Code	Purpose
Delete	`cursor.execute('DELETE FROM ...')`	Remove rows
Commit	`connection.commit()`	Save all changes
Close	`connection.close()`	Free resources

---

12. Python Logging

> **"Logging is a crucial aspect of any application, providing a way to track events, errors, and operational information. Python's built-in logging module offers a flexible framework for emitting log messages from Python programs."**

12.1 Importing the Logging Module

```
import logging
```

12.2 Log Levels

Python's `logging` module has several log levels indicating the **severity** of events. Listed from lowest to highest severity:

Level	Numeric Value	Description
`DEBUG`	10	Detailed information, typically of interest only when diagnosing problems
`INFO`	20	Confirmation that things are working as expected
`WARNING`	30	Something unexpected happened, or indicative of a problem in the near future (e.g. "disk space low"). Software is still working as expected
`ERROR`	40	Due to a more serious problem, the software has not been able to perform some function
`CRITICAL`	50	A very serious error, indicating that the program itself may be unable to continue running

> **Note:** When you set a level (e.g. `DEBUG`), all messages at **that level and above** will be shown. Setting `WARNING` means `DEBUG` and `INFO` messages will be suppressed.

12.3 Basic Logging Setup

```
# Configure the basic logging settings
logging.basicConfig(level=logging.DEBUG)  # There are other levels too

# Log messages with different severity levels
logging.debug("This is a debug message")
logging.info("This is an info message")
logging.warning("This is a warning message")
```

```
logging.error("This is an error message")
logging.critical("This is a critical message")
```

> **Result:**
>
> ```
> DEBUG:root:This is a debug message
> INFO:root:This is an info message
> WARNING:root:This is a warning message
> ERROR:root:This is an error message
> CRITICAL:root:This is a critical message
> ```

> **Note:** The format of each log line by default is `LEVEL:logger_name:message`. The logger name `root` is the default top-level logger.

12.4 Configuring Logging Format and Output File

You can customize how log messages look and where they are saved using `basicConfig()` with additional parameters.

```
logging.basicConfig(
    filename='app.log',                          # Save logs to this file
    filemode='w',                                # 'w' overwrites the file each
run; use 'a' to append
    level=logging.DEBUG,
    format='%(asctime)s - %(name)s - %(message)s',
    datefmt='%Y-%m-%d %H:%M:%S',
    force=True   # Required if basicConfig() was already called earlier in the
session
                 # (logging does not allow re-configuration without force=True)
)

# Checking the output
logging.debug("This is a debug message")
logging.info("This is an info message")
logging.warning("This is a warning message")
logging.error("This is an error message")
logging.critical("This is a critical message")
```

> `format` **placeholders:**
>
> ```
> %(asctime)s    — Timestamp of when the log was created
> %(name)s       — Name of the logger (default: root)
> %(levelname)s  — Log level name (DEBUG, INFO, WARNING, etc.)
> %(message)s    — The actual log message
> ```

> `datefmt` uses standard `strftime` codes, e.g. `'%Y-%m-%d %H:%M:%S'` → `2024-01-15 09:30:00`

12.5 Configuring Logging with Handlers

Handlers are the components that determine **where** your log messages actually go. You can attach multiple handlers to send logs to different destinations simultaneously.

```
logging.basicConfig(
    level=logging.DEBUG,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s',
    datefmt='%Y-%m-%d %H:%M:%S',
    handlers=[
        logging.FileHandler("app1.log"),   # Writes log messages to app1.log file
        logging.StreamHandler()            # Outputs log messages to the console
(stdout)
    ]
)
```

> **Common Handlers:**
>
> - `FileHandler(filename)` — writes logs to a file
> - `StreamHandler()` — writes logs to the terminal / console
> - Both can be used together so logs appear in the terminal **and** get saved to a file at the same time

---

12.6 Practical Example — Organizing Logs Into a Folder

It is good practice to keep your logs organized in a dedicated `logs/` folder. Here is an example project structure:

```
📦 Day 11
├ 📂 logs
│ ├ 📂 __pycache__
│ │ └ 📜 logger.cpython-314.pyc
│ ├ 📜 app.log
│ ├ 📜 logger.py       ← Central logger configuration
│ └ 📜 test.py         ← File that uses the logger
├ 📜 app.log
└ 📜 LoggingPracticalImplementation.ipynb
```

`logger.py` — Central Logger Configuration

Create a reusable logger in a dedicated file so any module in your project can import it:

```python
import logging
import os

# Ensure the logs directory exists
os.makedirs('logs', exist_ok=True)

logging.basicConfig(
    filename='logs/app.log',
    filemode='a',                    # Append mode — don't overwrite previous logs
    level=logging.DEBUG,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s',
    datefmt='%Y-%m-%d %H:%M:%S'
)

logger = logging.getLogger('MyAppLogger')
```

`test.py` — Using the Logger

```python
from logger import logger

logger.debug("App started")
logger.info("Loading data...")
logger.warning("Data file is missing a column")
logger.error("Failed to connect to database")
logger.critical("Application is shutting down")
```

Quick Reference — `basicConfig()` Parameters

Parameter	Purpose	Example
`level`	Minimum severity level to log	`logging.DEBUG`
`filename`	Save logs to a file	`'app.log'`
`filemode`	File write mode	`'w'` (overwrite) or `'a'` (append)
`format`	Structure of each log line	`'%(asctime)s - %(levelname)s - %(message)s'`
`datefmt`	Format for the timestamp	`'%Y-%m-%d %H:%M:%S'`
`handlers`	List of output destinations	`[FileHandler(...), StreamHandler()]`
`force`	Re-apply config even if already set	`True`

---

13. Logging With Multiple Loggers / Modules

> **"You can create multiple loggers for different parts of your application."**

Instead of using the default `root` logger for everything, named loggers let each module have its **own identity, severity level, and behavior** — making logs much easier to trace and filter in larger projects.

13.1 Creating Multiple Loggers

Each logger is created with `logging.getLogger("name")` and can have its own level set independently.

```
import logging

# Create a logger for module1
logger1 = logging.getLogger("module1")
logger1.setLevel(logging.DEBUG)      # Captures DEBUG and above

# Create a logger for module2
logger2 = logging.getLogger("module2")
logger2.setLevel(logging.WARNING)    # Captures WARNING and above only

# Configure shared logging format
logging.basicConfig(
    level=logging.DEBUG,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s',
    datefmt='%Y-%m-%d %H:%M:%S'
)
```

> **Note:** `basicConfig()` sets the global format and the root logger's level. Each named logger's own `.setLevel()` overrides this for that specific logger — so `logger2` will ignore `DEBUG` and `INFO` messages even though the root is set to `DEBUG`.

13.2 Logging Messages From Different Loggers

```
logger1.debug("This is a debug message for module 1")
logger2.warning("This is a warning message for module 2")
logger2.error("This is an error message for module 2")

# Now instead of 'root', the module name appears in each log line
```

> **Result:**

```
2026-03-18 09:54:33 - module1 - DEBUG   - This is a debug message for module
1
2026-03-18 09:54:33 - module2 - WARNING - This is a warning message for
module 2
2026-03-18 09:54:33 - module2 - ERROR   - This is an error message for
module 2
```

> **Notice:** `logger2.debug(...)` or `logger2.info(...)` would produce **no output** because `logger2`'s level is set to `WARNING` — anything below that is silently ignored.

13.3 How Named Loggers Work

Each call to `logging.getLogger("name")` returns the **same logger object** if the name has been used before — loggers are singletons identified by name. This means you can safely import and reuse the same logger across multiple files.

```python
# In any file across your project:
logger = logging.getLogger("module1")   # Returns the same logger1 object as above
```

Logger Naming Convention

It is best practice to name loggers after the module they live in using `__name__`:

```python
# Inside myapp/database.py
logger = logging.getLogger(__name__)
# __name__ resolves to 'myapp.database' automatically
```

This makes the logger name match the file path, so log output tells you exactly which file generated each message.

13.4 Practical Example — Multiple Loggers in a Project

```python
import logging

# Shared format
logging.basicConfig(
    level=logging.DEBUG,
    format='%(asctime)s - %(name)s - %(levelname)s - %(message)s',
    datefmt='%Y-%m-%d %H:%M:%S'
)

# Logger for data loading module — verbose, captures everything
data_logger = logging.getLogger("data_loader")
```

```
data_logger.setLevel(logging.DEBUG)

# Logger for model training module — only important messages
model_logger = logging.getLogger("model_trainer")
model_logger.setLevel(logging.WARNING)

# Logger for API module — errors only
api_logger = logging.getLogger("api")
api_logger.setLevel(logging.ERROR)

# Usage
data_logger.debug("Loading dataset from disk...")       # shown
data_logger.info("Dataset loaded: 1000 rows")           # shown
model_logger.debug("Starting epoch 1")                  # suppressed
model_logger.warning("Validation loss is increasing")   # shown
api_logger.info("Request received")                     # suppressed
api_logger.error("Failed to serialize response")        # shown
```

Result:

```
2026-03-18 09:54:33 - data_loader    - DEBUG   - Loading dataset from
disk...
2026-03-18 09:54:33 - data_loader    - INFO    - Dataset loaded: 1000 rows
2026-03-18 09:54:33 - model_trainer  - WARNING - Validation loss is
increasing
2026-03-18 09:54:33 - api            - ERROR   - Failed to serialize
response
```

Quick Reference — Single vs Multiple Loggers

	Root Logger	**Named Logger**
How to get	`logging.warning(...)`	`logging.getLogger("name")`
Logger name in output	`root`	Your custom name
Level control	Global only	Per-logger, independent
Best for	Quick scripts	Multi-module applications
Reusable across files	No	Yes — same name returns same object

---

14. Multithreading and Multiprocessing

14.1 Core Concepts — Program, Process, and Thread

Program

A **program** is a sequence of instructions written in a programming language — essentially a static file sitting on disk.

```
Example: Google Chrome → chrome.exe → a sequence of instructions
```

Process

A **process** is an instance of a program that is currently being **executed**. When you open Google Chrome, you create a process.

Every process contains:

- **Code Segment** — the program's instructions
- **Data Segment** — global and static variables
- **Heap** — dynamically allocated memory
- **Stack** — function calls and local variables
- **Registers** — CPU state

> **Key properties of processes:**
>
> - Every process gets its **own separate memory space**
> - One process **cannot corrupt** another process's memory
> - Running more processes requires more I/O resources, which increases the time to switch between them (context switching overhead)

Thread

A **thread** is a unit of execution **within** a process. A single process can have one or many threads running inside it.

Single-Threaded Process

Has one thread. The thread shares the process's code segment, data segment, and heap, but has its own **stack** and **registers**. Think of it as a sub-process.

Multi-Threaded Process

Every thread in the process has its **own stack and registers**, but all threads **share** the code segment, data segment, and heap within the same process.

	Code Segment	Data Segment	Heap	Stack	Registers
Process	✅ owns	✅ owns	✅ owns	✅ owns	✅ owns
Each Thread	shared	shared	shared	✅ own	✅ own

14.2 Multithreading in Python

When to Use Multithreading

Multithreading is best suited for:

- **I/O-Bound Tasks** — tasks that spend most of their time *waiting* for I/O operations (e.g. file reads, network requests, database queries). While one thread is waiting, another can run.
- **Concurrent Execution** — when you want to improve throughput by performing multiple operations at the same time.

> **Note:** Due to Python's **GIL (Global Interpreter Lock)**, multithreading does **not** speed up CPU-bound tasks (heavy computation). For CPU-bound work, use **multiprocessing** instead.

14.2.1 Without Multithreading (Sequential)

Running two functions one after the other on a single thread — each `sleep(2)` fully blocks the program:

```python
import threading
import time

def print_numbers():
    for i in range(5):
        time.sleep(2)
        print(f"Number: {i}")

def print_letters():
    for letter in "abcde":
        time.sleep(2)
        print(f"Letter: {letter}")

# Before Multithreading — sequential execution
start_time = time.time()
print_letters()
print_numbers()
finished_time = time.time() - start_time
print(f"Finished in {finished_time:.4f} seconds")
```

> **Result:**
>
> ```
> Letter: a
> Letter: b
> ```

```
Letter: c
Letter: d
Letter: e
Number: 0
Number: 1
Number: 2
Number: 3
Number: 4
Finished in 20.0080 seconds
```

Here, we run two functions sequentially on a single thread. Each function includes `sleep(2)`, so while one function is sleeping the program does not execute the other function. The total runtime is roughly the **sum** of both functions (5 × 2s + 5 × 2s = 20s).

14.2.2 With Multithreading (Concurrent)

With multithreading, the two functions run in **separate threads** within the same process. When one thread is sleeping (waiting on I/O), the other thread keeps running — they overlap in time instead of waiting for each other.

```python
import threading
import time

def print_numbers():
    for i in range(5):
        time.sleep(2)
        print(f"Number: {i}")

def print_letters():
    for letter in "abcde":
        time.sleep(2)
        print(f"Letter: {letter}")

# Creating 2 threads
t1 = threading.Thread(target=print_numbers)
t2 = threading.Thread(target=print_letters)

# After Multithreading — concurrent execution
start_time = time.time()
t1.start()    # Start thread 1
t2.start()    # Start thread 2

# Wait for both threads to finish before moving on
t1.join()
t2.join()

finished_time = time.time() - start_time
print(f"Finished in {finished_time:.4f} seconds")
```

> **Result:**
>
> ```
> Number: 0
> Letter: a
> Number: 1  Letter: b
> Letter: c
> Number: 2
> Letter: d
> Number: 3
> Letter: e  Number: 4
> Finished in 10.0183 seconds
> ```

> **We save 10 seconds by using multithreading.** Both functions now run simultaneously — the total time is the length of the *longer* function, not the sum of both.

> **Notice:** The output order of numbers and letters is interleaved and slightly unpredictable — that is normal with threads. Both threads are racing to print at the same time, so their output can appear on the same line.

14.3 Key Threading Methods

Method / Class	Purpose
`threading.Thread(target=fn)`	Create a new thread that will run `fn`
`t.start()`	Start the thread — begins executing `fn`
`t.join()`	Block the main thread until `t` finishes
`threading.current_thread()`	Returns the currently running thread object
`threading.active_count()`	Returns the number of threads currently alive

14.4 Sequential vs Multithreaded — Summary

	Sequential (Single Thread)	Multithreaded
Execution	One task at a time	Multiple tasks overlap
Total time (I/O-bound)	Sum of all tasks	~Time of the longest task
Output order	Predictable	Interleaved / unpredictable
Best for	Simple scripts, CPU-bound work	I/O-bound tasks, concurrent operations
Risk	None	Race conditions if threads share data

14.5 Multiprocessing

14.5.1 When to Use Multiprocessing

Multiprocessing is best suited for:

- **CPU-Bound Tasks** — tasks that are heavy on CPU usage (e.g. mathematical computations, data processing, image manipulation)
- **Parallel Execution** — taking advantage of **multiple CPU cores** to run tasks truly simultaneously, not just concurrently

> **Multithreading vs Multiprocessing:**

	Multithreading	Multiprocessing
Best for	I/O-bound tasks (waiting on files, network)	CPU-bound tasks (heavy computation)
Runs on	Multiple threads, one process	Multiple separate processes
Memory	Shared between threads	Each process gets its own memory
GIL limitation	Yes — threads can't truly run in parallel	No — each process bypasses the GIL
Speed gain	When waiting/sleeping	When doing heavy computation

---

14.5.2 Multiprocessing Example

> **Important:** Multiprocessing code **must** be wrapped inside `if __name__ == "__main__":` — this prevents infinite spawning of child processes when the script is imported or re-run by each worker.

```
import multiprocessing
import time

def square_numbers():
    for i in range(5):
        time.sleep(1)
        print(f"Square: {i*i}")

def cube_numbers():
    for i in range(5):
        time.sleep(1.5)
        print(f"Cube: {i*i*i}")

if __name__ == "__main__":

    # Create 2 processes
    p1 = multiprocessing.Process(target=square_numbers)
```

```
    p2 = multiprocessing.Process(target=cube_numbers)
    t = time.time()

    # Start the processes
    p1.start()
    p2.start()

    # Wait for both processes to complete
    p1.join()
    p2.join()

    finished_time = time.time() - t
    print(finished_time)
```

> **Note:** Multiprocessing cannot be run directly inside a Jupyter Notebook cell — it must be executed from a separate `.py` file. The output below is from running `multiprocessing_example.py` in the terminal.

> **Result (executed in separate file):**
>
> ```
> Square: 0
> Cube: 0
> Square: 1
> Square: 4
> Cube: 1
> Square: 9
> Cube: 8
> Square: 16
> Cube: 27
> Cube: 64
> 7.582595586776733
> ```

> **Why ~7.5 seconds?** The two processes run in parallel on separate CPU cores, so the total time is driven by the **slower** process. `cube_numbers` does 5 iterations × 1.5s sleep = 7.5s. Compare this to sequential execution which would take 5×1 + 5×1.5 = **12.5 seconds**.

14.5.3 Key Multiprocessing Methods

Method / Class	Purpose
`multiprocessing.Process(target=fn)`	Create a new process that will run `fn`
`p.start()`	Spawn and start the process
`p.join()`	Block the main process until `p` finishes
`p.is_alive()`	Returns `True` if the process is still running
`multiprocessing.cpu_count()`	Returns the number of CPU cores available

---

15. Using Pool Executor for Multithreading and Multiprocessing

> **"Instead of manually creating, starting, and joining individual threads or processes (which is tedious and resource-heavy), you create a pool of workers that stay alive and check out tasks from a queue as they become available."**

The `concurrent.futures` module provides a high-level, unified interface for both multithreading and multiprocessing through two executor classes: `ThreadPoolExecutor` and `ProcessPoolExecutor`.

15.1 Multithreading With `ThreadPoolExecutor`

15.1.1 Basic Example

```python
from concurrent.futures import ThreadPoolExecutor
import time

def print_number(number):
    time.sleep(1)
    return f"Number: {number}"

numbers = [1, 2, 3, 4, 5]

with ThreadPoolExecutor(max_workers=3) as executor:
    results = executor.map(print_number, numbers)

for result in results:
    print(result)
```

> **Result:**
>
> ```
> Number: 1
> Number: 2
> Number: 3
> Number: 4
> Number: 5
> ```

15.1.2 Why Is This More Efficient?

Think of it like a grocery store checkout:

- **Sequential (normal for-loop):** One cashier scans each item one by one. Each item takes 1 second → total time = **5 seconds**.

- `ThreadPoolExecutor(max_workers=3)`**:** The store opens 3 checkout lanes simultaneously.
  - Items 1, 2, 3 are scanned in parallel at t=0
  - At t=1s, those 3 are done → items 4 and 5 jump into the now-empty lanes
  - At t=2s, everything is finished → total time = **~2 seconds**

The math:

- Sequential: 5 tasks × 1 second = **5 seconds**
- With ThreadPoolExecutor (3 workers): **~2 seconds**

15.1.3 Breaking Down the Syntax

```
with ThreadPoolExecutor(max_workers=3) as executor:
```

This is a **context manager**.

- `ThreadPoolExecutor(max_workers=3)` — initializes the pool and tells Python to keep a maximum of **3 threads** working at any given time
- `with ... as executor:` — the safety net. As soon as the code block finishes, all threads are automatically cleaned up and shut down. Without `with`, you would have to manually call `executor.shutdown()`

```
results = executor.map(print_number, numbers)
```

This is the high-level way to distribute work across the pool.

- Works exactly like Python's built-in `map(function, list)` — takes every item in `numbers` and feeds it into `print_number`
- **Unlike** a regular `map`, it doesn't wait for item 1 to finish before starting item 2 — it hands items out to available workers as fast as possible
- **Order is preserved** — `executor.map` always returns results in the original input order, even if a later task finishes before an earlier one

---

15.1.4 When Would Multithreading NOT Be Efficient?

If you replace `time.sleep(1)` with a heavy CPU computation (like `number * 9999999`), `ThreadPoolExecutor` may not speed things up. This is because of Python's **GIL (Global Interpreter Lock)** — standard Python threads are great at *waiting* (I/O-bound tasks like downloading files or sleeping) but struggle to do heavy *computation* simultaneously.

> **Rule of thumb:**
>
> - I/O-bound (sleeping, network calls, file reads) → use `ThreadPoolExecutor`
> - CPU-bound (heavy math, data processing) → use `ProcessPoolExecutor`

15.2 Multiprocessing With `ProcessPoolExecutor`

15.2.1 How It Differs From ThreadPoolExecutor

`ProcessPoolExecutor` bypasses the GIL entirely by spawning **entirely separate Python processes**, each with its own memory space and its own CPU core. This allows truly parallel computation — multiple tasks running at the **exact same instant** on different cores.

```
ThreadPoolExecutor  → multiple threads  → one process  → shares memory  → GIL
applies
ProcessPoolExecutor → multiple processes → own memory each → own CPU core → GIL
bypassed
```

15.2.2 Basic Example

```python
from concurrent.futures import ProcessPoolExecutor
import time

def square_number(number):
    time.sleep(1)
    return f"Square: {number*number}"

numbers = [1, 2, 3, 4, 5]

with ProcessPoolExecutor(max_workers=3) as executor:
    results = executor.map(square_number, numbers)

for result in results:
    print(result)
```

> **Note:** Like raw `multiprocessing`, `ProcessPoolExecutor` must be run from a `.py` file, not directly in a Jupyter Notebook cell.

> **Result (executed in separate file):**
>
> ```
> Square: 1
> Square: 4
> Square: 9
> Square: 16
> Square: 25
> ```

15.3 `ThreadPoolExecutor` vs `ProcessPoolExecutor` — Full Comparison

	`ThreadPoolExecutor`	`ProcessPoolExecutor`
Import	`concurrent.futures`	`concurrent.futures`
Syntax	Identical	Identical

	`ThreadPoolExecutor`	`ProcessPoolExecutor`
Worker type	Threads	Separate processes
Memory	Shared	Isolated per process
GIL	Affected	Bypassed
Best for	I/O-bound tasks	CPU-bound tasks
Jupyter Notebook	Works	Must run in `.py` file
Overhead	Low	Higher (process startup cost)
Result order	Preserved by `map`	Preserved by `map`

15.4 Quick Reference — `executor.map` vs Manual Threading

	Manual `threading.Thread`	`executor.map`
Setup	Create, start, join each thread manually	One line
Worker limit	No built-in limit	Controlled by `max_workers`
Result collection	Manual (use queues or shared lists)	Returned directly, in order
Cleanup	Manual `join()` for each thread	Automatic via `with` block
Code complexity	High	Low

15.5 Real-World Practical Examples

15.5.1 Multithreading for I/O-Bound Tasks — Web Scraping

Scenario: Web scraping involves making numerous network requests to fetch web pages. These tasks are **I/O-bound** because they spend most of their time waiting for responses from servers. Multithreading significantly improves performance by allowing multiple pages to be fetched **concurrently** — while one thread waits for a server response, others keep working.

```
import threading
import requests
from bs4 import BeautifulSoup

urls = [
    'https://docs.langchain.com/oss/python/deepagents/overview',
    'https://docs.langchain.com/oss/python/langchain/overview',
    'https://docs.langchain.com/oss/python/langgraph/overview'
]

def fetch_content(url):
    response = requests.get(url)
    soup = BeautifulSoup(response.content, 'html.parser')
    print(f"Fetched {len(soup.text)} characters from {url}")
```

```
threads = []

for url in urls:
    thread = threading.Thread(target=fetch_content, args=(url,))
    threads.append(thread)
    thread.start()

for thread in threads:
    thread.join()

print("All web pages fetched...")
```

> **How it works:**
>
> - A new thread is created for each URL and started immediately — all 3 requests go out nearly simultaneously
> - `thread.join()` in the second loop ensures the main program waits for all threads to finish before printing the final message
> - Without multithreading, the 3 requests would fire one after the other — total time = sum of all response times. With multithreading, total time ≈ the slowest single request

> **Why not multiprocessing here?** Network requests spend almost all their time *waiting* — the CPU is barely involved. Spinning up separate processes would add overhead with no benefit. Threads are the right tool for this job.

15.5.2 Multiprocessing for CPU-Bound Tasks — Factorial Calculation

Scenario: Factorial calculations for large numbers involve significant computational work — there is no waiting, just pure CPU crunching. Multiprocessing distributes the workload across **multiple CPU cores**, running calculations truly in parallel and cutting down overall time.

```
import math
import sys
import time
import multiprocessing

sys.set_int_max_str_digits(100000)   # Allows printing very large integers without
truncation

def compute_factorial(number):
    print(f"Computing factorial of {number}")
    result = math.factorial(number)
    print(f"Factorial of {number} computed")
    return result

if __name__ == "__main__":
    numbers = [5000, 6000, 700, 8000]
```

```
    start_time = time.time()

    # Create a pool of worker processes — one per CPU core by default
    with multiprocessing.Pool() as pool:
        results = pool.map(compute_factorial, numbers)

    end_time = time.time()

    print(f"Results: {results}")
    print(f"Time taken: {end_time - start_time} seconds")
```

> **Key points:**
>
> - `multiprocessing.Pool()` creates a pool of worker processes equal to the number of CPU cores available on your machine (you can also pass an explicit number e.g. `Pool(4)`)
> - `pool.map(compute_factorial, numbers)` distributes each number in the list to a different worker process — each core handles one factorial at the same time
> - `sys.set_int_max_str_digits(100000)` is needed because Python limits how many digits it will print for very large integers by default — factorials of 5000+ have thousands of digits
> - Must be run from a `.py` file (not a Jupyter Notebook cell) because of the `if __name__ == "__main__":` requirement

> **Why not multithreading here?** Factorial computation is pure CPU work — no waiting involved. Python's GIL would prevent threads from running on separate cores simultaneously. Separate processes each get their own core and their own GIL, so they can truly run in parallel.

---

16. Python Memory Management

> **"Memory management in Python involves a combination of automatic garbage collection, reference counting, and various internal optimizations to efficiently manage memory allocation and deallocation. Understanding these mechanisms can help developers write more efficient and robust applications."**

Key Concepts Covered:

- Memory Allocation and Deallocation
- Reference Counting
- Garbage Collection
- The `gc` Module
- Memory Management Best Practices

16.1 Reference Counting

Reference counting is the primary method Python uses to manage memory. Every object in Python maintains an internal count of how many references are pointing to it. When that count drops to **zero**, the memory occupied by the object is automatically deallocated.

Use `sys.getrefcount(obj)` to inspect the reference count of any object.

> **Note:** `getrefcount()` itself creates a temporary reference, so the count is always **+1** higher than you might expect.

Example — Watching the Reference Count Change

```
import sys

a = []
# 2 references: one from 'a', one from getrefcount() itself
print(sys.getrefcount(a))
```

> **Result:**
>
> ```
> 2
> ```

```
b = a
# 3 references: 'a', 'b', and getrefcount()
print(sys.getrefcount(b))
```

> **Result:**
>
> ```
> 3
> ```

```
del b
# Back to 2: 'b' was deleted, only 'a' and getrefcount() remain
print(sys.getrefcount(a))
```

> **Result:**
>
> ```
> 2
> ```

> **Key takeaway:** When the reference count drops to **0**, Python immediately frees the memory occupied by that object — no manual cleanup needed.

16.2 Garbage Collection

Reference counting works well in most cases, but it has one weakness: **reference cycles**. A reference cycle occurs when two or more objects reference each other — their counts never reach zero even though nothing else in the program can reach them.

> **Example of a cycle:** `obj1.ref = obj2` and `obj2.ref = obj1` — even after `del obj1` and `del obj2`, both objects still point to each other, so their counts never hit 0.

Python solves this with a **cyclic garbage collector** (unlike some other languages that rely on destructors). Python's `gc` module handles detecting and cleaning up these unreachable cycles.

16.2.1 The `gc` Module

```
import gc

# Enable garbage collection (on by default)
gc.enable()

# Disable garbage collection
gc.disable()

# Manually trigger garbage collection
# Returns the number of unreachable objects collected
gc.collect()
```

> **Result of `gc.collect()`:**

```
0
```

```
# Get per-generation garbage collection statistics
print(gc.get_stats())
```

Result:

```
[
  {'collections': 0,  'collected': 0,    'uncollectable': 0},
  {'collections': 12, 'collected': 498,  'uncollectable': 0},
  {'collections': 2,  'collected': 3048, 'uncollectable': 0}
]
```

What are generations? Python's GC uses a **generational** approach with 3 generations (0, 1, 2). New objects start in generation 0. If they survive a collection cycle, they move up to generation 1, then 2. Older generations are checked less frequently — this is efficient because most short-lived objects die young.

```
# Inspect objects that could not be collected (uncollectable garbage)
print(gc.garbage)
```

Result:

```
[]
```

16.2.2 Handling Circular References

```
import gc

class MyObject:
    def __init__(self, name):
        self.name = name
        print(f"Object {self.name} created")

    def __del__(self):
        print(f"Object {self.name} deleted")

# Creating a circular reference — bad practice, leads to memory leaks
obj1 = MyObject("obj1")
obj2 = MyObject("obj2")
```

```
obj1.ref = obj2    # obj1 points to obj2
obj2.ref = obj1    # obj2 points back to obj1

# Deleting our variables — but the cycle keeps both objects alive!
del obj1
del obj2

# At this point the objects are NOT yet freed because of the cycle.
# We must manually trigger the garbage collector to break the cycle.
gc.collect()
```

Result:

```
Object obj1 created
Object obj2 created
Object obj1 deleted
Object obj2 deleted
```

Note: After `del obj1` and `del obj2`, the `__del__` methods are NOT called immediately — Python can't free them because their reference counts are still > 0 due to the cycle. Only after `gc.collect()` are they properly detected and freed.

16.3 Memory Management Best Practices

Practice	Why It Matters
Use local variables	Local variables have a shorter lifespan and are freed sooner than global variables
Avoid circular references	Cycles prevent reference counting from working — require GC to clean up
Use generators	Produce items one at a time — only one item lives in memory at once
Explicitly delete objects	Use `del` to free large objects proactively rather than waiting for GC
Profile memory usage	Use `tracemalloc` or `memory_profiler` to find leaks and hotspots

Why These Matter

Generators vs Lists: If you're processing a million rows of data, a list loads all of them into RAM at once — which might crash your app. A generator acts like a **conveyor belt**, bringing you one item at a time so memory usage stays flat.

The power of `tracemalloc`: It acts like a detective — it can tell you exactly which line of code is responsible for allocating the most memory, helping you find hidden leaks caused by circular references or large forgotten objects.

The `del` statement: While Python is great at cleaning up after itself, using `del` is like proactively taking out the trash rather than waiting for the garbage truck. It's particularly useful for large datasets or image processing buffers.

16.4 Generators and the `yield` Keyword

What Is `yield`?

The `yield` keyword turns a regular function into a **generator function**. Instead of computing all values at once and returning them in a list, a generator produces values **lazily** — one at a time, on demand.

Key behaviours of `yield`:

- The function's code is **not executed** when first called — it only runs when iterated
- Execution is divided into steps, **one step per `yield`**
- Unlike `return` (which stops the function entirely), `yield` **pauses** execution and resumes from where it left off on the next iteration
- The return value of calling the function is an **iterator** — not a list

Example — Memory-Efficient Number Generator

```
def generate_numbers(n):
    for i in range(n):
        yield i          # Produces one number at a time, pauses here

# Using the generator — even for 100,000 numbers, only ONE integer is in memory at
a time
for num in generate_numbers(100000):
    print(num)
    if num > 10:
        break
```

Result:

```
0
1
2
3
4
5
6
7
8
9
10
11
```

Memory comparison:

```
# List — loads ALL 100,000 integers into RAM immediately
numbers = [i for i in range(100000)]

# Generator — holds only the current integer in memory
numbers = generate_numbers(100000)
```

16.5 Profiling Memory Usage With `tracemalloc`

`tracemalloc` is a **built-in** Python library that traces memory allocations. It tells you exactly which lines of your code are using the most memory — essential for tracking down leaks and optimizing large applications.

```
import tracemalloc

def create_list():
    return [i for i in range(10000)]

def main():
    tracemalloc.start()         # Begin tracing memory allocations

    create_list()

    snapshot = tracemalloc.take_snapshot()              # Capture current memory
state
    top_stats = snapshot.statistics('lineno')           # Sort by line number

    print("[ Top 10 Memory Consumers ]")
    for stat in top_stats[:10]:
        print(stat)

main()
```

Result (top entries — full output includes all allocations across the entire Python environment):

```
[ Top 10 Memory Consumers ]
inspect.py:2178:        size=304 KiB, count=11,  average=27.7 KiB
selectors.py:305:       size=144 KiB, count=3,   average=48.0 KiB
compilerop.py:174:      size=19.2 KiB, count=203, average=97 B
html.py:514:            size=18.1 KiB, count=228, average=82 B
inspect.py:2199:        size=13.8 KiB, count=40,  average=353 B
tracemalloc.py:193:     size=9408 B,   count=168, average=56 B
html.py:511:            size=6600 B,   count=3,   average=2200 B
socket.py:806:          size=6448 B,   count=124, average=52 B
html.py:835:            size=5814 B,   count=126, average=46 B
html.py:868:            size=5568 B,   count=60,  average=93 B
```

> **Reading the output:** Each line shows `file:line_number: size=X, count=Y, average=Z`.
>
> - `size` — total memory allocated from that line
> - `count` — number of individual allocations
> - `average` — average size per allocation
>
> In a real project, you would filter for your own file paths to focus on your code's allocations rather than library internals.

`tracemalloc` Key Functions

Function	Purpose
`tracemalloc.start()`	Begin recording memory allocations
`tracemalloc.take_snapshot()`	Capture a snapshot of current allocations
`snapshot.statistics('lineno')`	Get allocations sorted by line number
`snapshot.statistics('filename')`	Get allocations sorted by file
`tracemalloc.stop()`	Stop tracing
`tracemalloc.get_traced_memory()`	Returns `(current, peak)` memory in bytes

16.6 Summary — Python Memory Management Flow

```
Object created
      │
      ▼
Reference count = 1
      │
   +──┴──────────────────────+
   │  New reference added │  → count++
   │  Reference deleted   │  → count--
   +──────────────────────+
      │
   count == 0?
   ┌──┴──┐
  YES    NO
   │      └──→ Object stays alive
   ▼
Memory freed immediately
(unless part of a cycle)
      │
   Cycle?
   ┌──┴──┐
  YES    NO
   │      └──→ Already freed above
   ▼
Cyclic GC detects
unreachable group
      │
      ▼
gc.collect() frees them
```

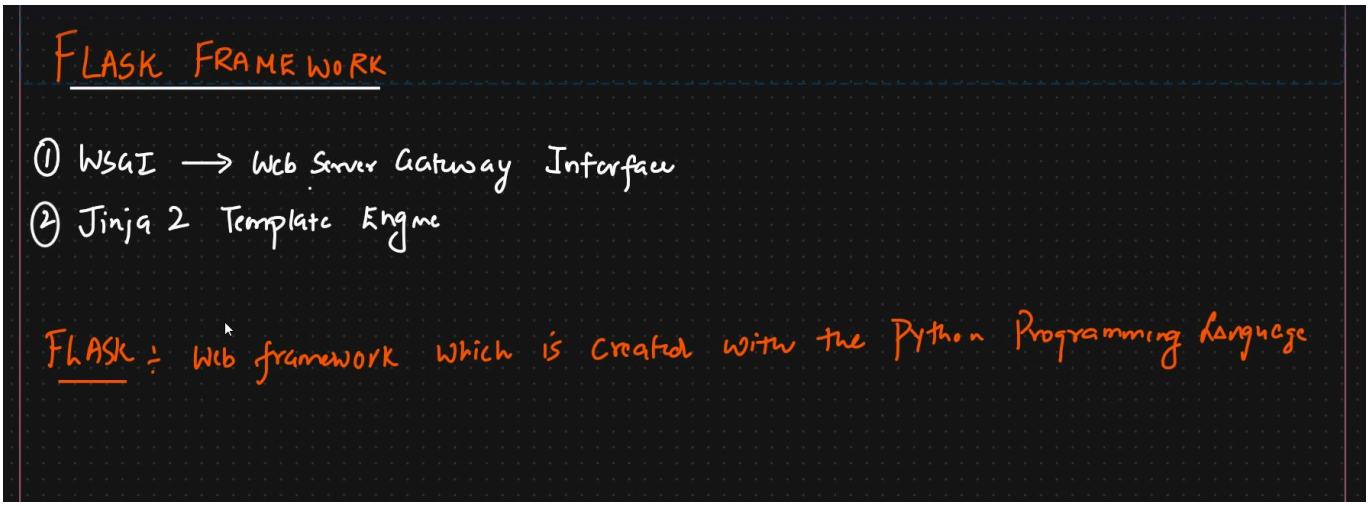
17. Flask Framework

> **"Flask is a web framework which is created with the Python programming language."**

Flask is a **lightweight, micro web framework** for Python. It is called "micro" because it keeps the core simple but extensible — it does not include built-in tools like a database layer or form validation, giving you the freedom to choose your own tools and libraries.

Flask is built on two core components:

1. **WSGI** — Web Server Gateway Interface
2. **Jinja2** — Template Engine



17.1 WSGI — Web Server Gateway Interface

WSGI (pronounced "wiz-gee") is a **protocol** — a specification that defines how a web server communicates with a Python web application.

Without WSGI, every web server (Apache, Nginx, IIS) would need to know the specific internal details of every Python framework (Flask, Django, FastAPI). WSGI solves this by defining a **single standard interface** between the two sides, so any WSGI-compatible server can work with any WSGI-compatible Python app.

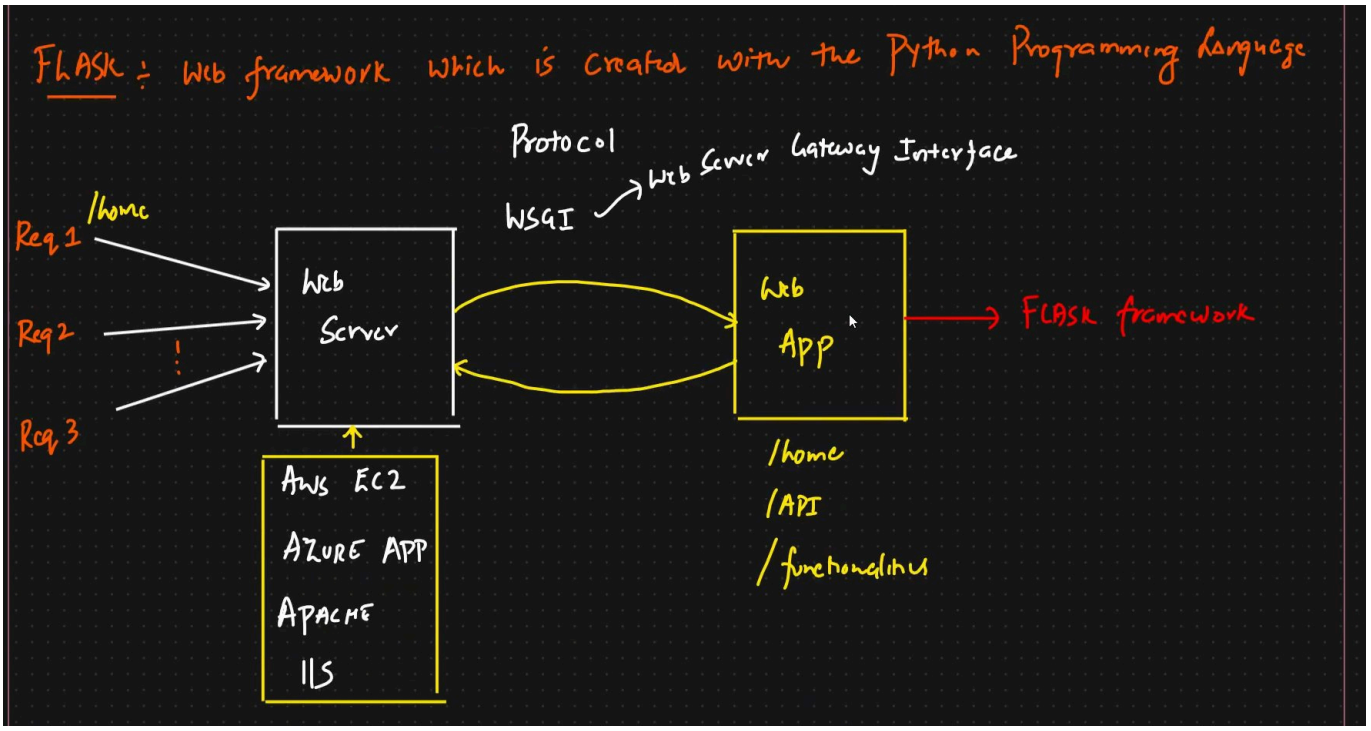
How It Works — Request Flow

```
Client Requests
      │
      ├── Req 1  (/home)
      ├── Req 2
      └── Req 3
            │
            ▼
     ┌──────────────┐
     │  Web Server  │  ◄── Receives raw HTTP requests
     │  (handles    │       and serves static files
     │   traffic)   │
     └──────────────┘
            │
            │  WSGI Protocol (the bridge)
            │  ◄──────────────────────────►
            ▼
     ┌──────────────┐
     │   Web App    │  ◄── Flask handles the routes:
     │  (Flask)     │       /home
     └──────────────┘       /api
                            /functionalinas
```

Web Server Options (WSGI-Compatible)

Web Server	Common Use
AWS EC2	Cloud-hosted server on Amazon Web Services
Azure App	Cloud-hosted server on Microsoft Azure
Apache	Popular open-source web server
IIS	Internet Information Services — Microsoft's web server

> **In development**, Flask runs its own built-in development server so you don't need Apache or Nginx. In **production**, you deploy behind a real WSGI server like Gunicorn or uWSGI, which sits in front of your Flask app.



17.2 Jinja2 — Template Engine

Jinja2 is a **web template engine** for Python. It allows you to write HTML pages that are dynamically populated with real data at runtime — so instead of writing a separate HTML file for every user or every result, you write one **template** and Jinja2 fills it in.

How It Works

```
Data Source  ──────────────►  Web Templates  ──►  Web Page (rendered HTML)
(backend)                      (HTML + Jinja2
                                syntax)
```

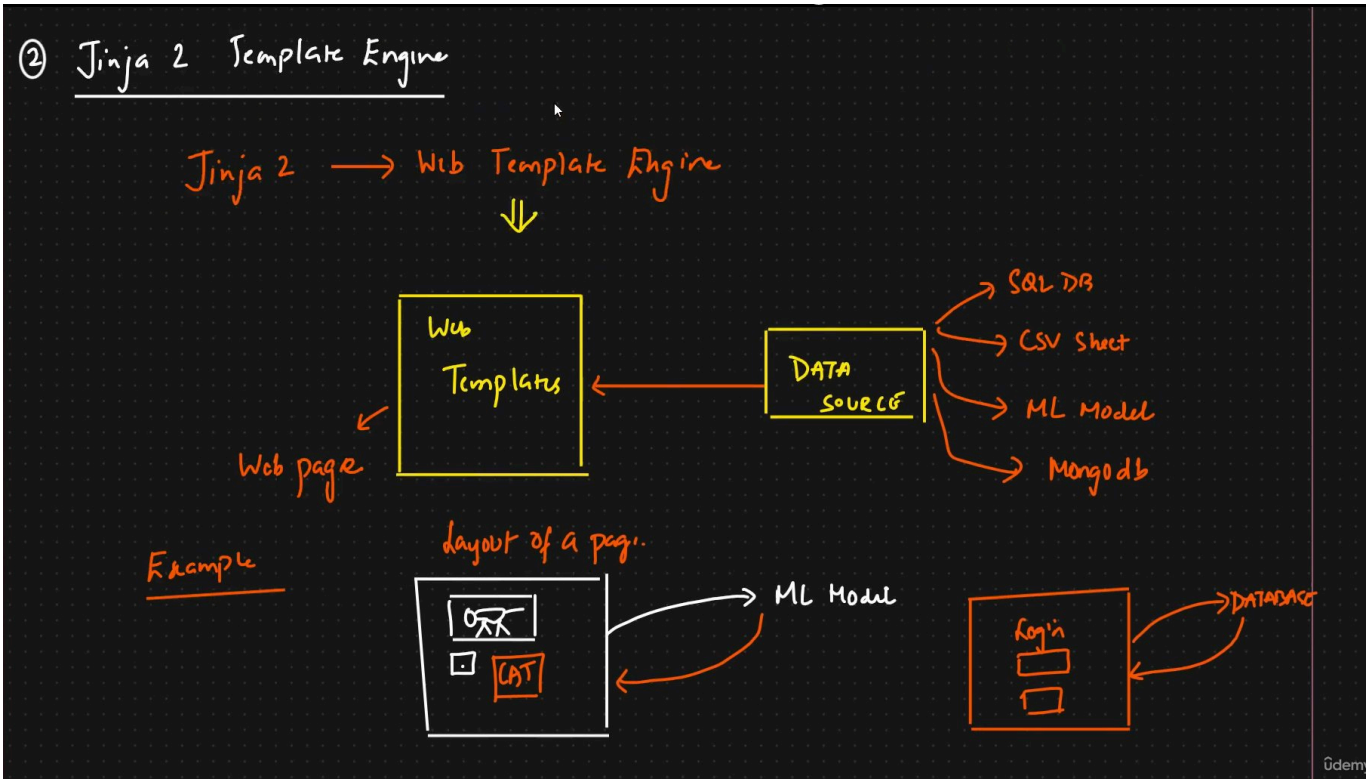
Data Sources Jinja2 Can Pull From

Source	Example
SQL Database	PostgreSQL, MySQL, SQLite
CSV Sheet	Pandas DataFrames, flat files
ML Model	Predictions, classification results
MongoDB	NoSQL document store

Real-World Page Examples

ML Prediction Page — A template with an image upload area and a text result box. The layout stays the same; the prediction result (e.g. "Cat") is injected dynamically by the ML model each time a user submits a photo.

Login Page — A template with username and password fields. The same HTML layout is reused for every user; whether the login succeeds is determined by checking the database dynamically.



17.3 Jinja2 Template Syntax

Jinja2 extends plain HTML with special tags:

Syntax	Purpose	Example
`{{ variable }}`	Output a variable value	`{{ username }}`
`{% if ... %}`	Conditional logic	`{% if user.is_admin %}`
`{% for ... %}`	Loop over a list	`{% for item in items %}`
`{% extends "base.html" %}`	Inherit a base template layout	Avoids repeating header/footer
`{% block content %}`	Define a replaceable section	Used with `extends`
`{{- -}}`	Strip whitespace	Cleaner output

Example — Dynamic HTML Template

```
<!-- templates/index.html -->
<!DOCTYPE html>
<html>
<body>
    <h1>Hello, {{ name }}!</h1>

    {% if items %}
```

```
        <ul>
        {% for item in items %}
            <li>{{ item }}</li>
        {% endfor %}
        </ul>
    {% else %}
        <p>No items found.</p>
    {% endif %}
</body>
</html>
```

```
# app.py — Flask passes data into the template
from flask import Flask, render_template

app = Flask(__name__)

@app.route('/')
def home():
    return render_template('index.html', name='Sher', items=['Python', 'Flask',
'ML'])
```

17.4 Flask — Minimal Working App

```
from flask import Flask

app = Flask(__name__)   # Creates the Flask application instance

@app.route('/')         # Decorator maps the URL '/' to this function
def home():
    return 'Hello, World!'

@app.route('/api')
def api():
    return '{"status": "ok"}'

if __name__ == '__main__':
    app.run(debug=True)  # Starts the built-in development server
                         # debug=True enables auto-reload on code changes
```

> **How routing works:** The `@app.route('/path')` decorator tells Flask — "when a request comes in for this URL, run this function and return its response." This maps directly to the `/home`, `/api`, `/functionalinas` routes shown in the architecture diagram.

17.5 Flask Architecture — Full Picture

```
                    BROWSER / CLIENT
                          │
              ┌───────────┼───────────┐
          Req 1        Req 2       Req 3
              └───────────┼───────────┘
                          ▼
             ┌────────────────────┐
             │     Web Server     │
             │  (Gunicorn/Apache/ │
             │   AWS EC2/Azure)   │
             └────────────────────┘
                          │  WSGI Protocol
                          ▼
             ┌────────────────────┐
             │    Flask App       │
             │                    │
             │  @route('/home')   │
             │  @route('/api')    │
             │  @route('/...')    │
             └────────────────────┘
                          │
                          │
             ┌────────────┼───────┐
             │                    │
       Jinja2 Templates         Business Logic
       (HTML rendering)         (Python functions,
             │                   DB queries, ML)
             ▼                      │
         Web Page              Data Sources
        (sent back             (SQL, CSV, ML
         to client)             Model, MongoDB)
```

17.6 Quick Reference — Flask Core Concepts

Concept	What It Is	Why It Matters
Flask	Python micro web framework	Lightweight, flexible, easy to get started
WSGI	Protocol between web server and Python app	Makes Flask deployable on any standard server
Jinja2	HTML template engine	Renders dynamic pages without duplicating HTML
Route	URL mapped to a Python function	Defines what happens when a URL is visited
`render_template()`	Loads and fills a Jinja2 template	Connects Python data to the HTML front-end
`debug=True`	Dev server auto-reloads on changes	Speeds up development — never use in production

---

17.7 Understanding Simple Flask App Skeleton

Flask is a lightweight **WSGI web framework** in Python. Every Flask application starts with a basic skeleton.

Key Concepts

- **WSGI** *(Web Server Gateway Interface)* — A standard interface between web servers and Python web applications.
- `Flask(__name__)` — Creates an instance of the Flask class. The `__name__` parameter tells Flask where to look for the entry point of the application.
- `if __name__ == "__main__"` — This block ensures the app runs only when the script is executed directly (not when imported).

Code

```
from flask import Flask

# WSGI Application
app = Flask(__name__)
# Creates an instance of the Flask class.
# "__name__" is passed so Flask knows the entry point of the application.

if __name__ == "__main__":  # Execution starts from here
    app.run()
```

> **Note:** This is the **basic skeleton** of any Flask web application. Everything else builds on top of this structure.

17.8 Adding Routes to the Basic Skeleton

Routes define the URLs of your web application and map them to Python functions.

Key Concepts

- `@app.route("/")` — A **decorator** that binds a URL path to a function. The `"/"` path represents the **home page**.
- `debug=True` — Enables **auto-restart** of the development server whenever code is updated. Without it, the server must be restarted manually after every change.
- `app.run()` — Starts the **local development server**. Should **not** be used in production.

Code

```
from flask import Flask

app = Flask(__name__)

# Home page route — equivalent to "google.com/"
@app.route("/")
def welcome():
    return "Welcome to this Flask App"

# Index page route
@app.route("/index")
def index():
    return "Welcome to the index page"

if __name__ == "__main__":
    app.run(debug=True)
    # debug=True → server auto-restarts on code changes
```

`app.run()` Parameters

Parameter	Type	Description
`host`	`str` | `None`	Hostname to listen on. Use `"0.0.0.0"` to make it externally visible.
`port`	`int` | `None`	Port number for the server.
`debug`	`bool` | `None`	Enables auto-reload and debugger.
`load_dotenv`	`bool`	Loads environment variables from `.env` file. Default: `True`.

Running the App

```
python ./routes.py
```

Result

```
* Running on http://127.0.0.1:5000
* Debug mode: on
* Restarting with stat
```

> ⚠️ **Warning:** `app.run()` is for **development only**. It is not intended to meet the security and performance requirements of a production server. For production deployment, use a proper **WSGI server** (e.g., Gunicorn, uWSGI).

Clean Version — Basic Skeleton + Routes (No Comments)

```python
from flask import Flask

app = Flask(__name__)

@app.route("/")
def welcome():
    return "Welcome to this Flask App"

@app.route("/index")
def index():
    return "Welcome to the index page"

if __name__ == "__main__":
    app.run(debug=True)
```